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Analysis of Feature Importance in a Neural Network for Predicting the Remaining Useful Life of Aircraft Engines

Master's Thesis
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List of Symbols

Symbol	Unit	Description
Latin symbols		
d		Number of input features / input dimensionality
f		Trained predictive model, $f : \mathbb{R}^d \to \mathbb{R}$
g		Local surrogate model (LIME)
k		Convolutional kernel size
m		Number of integration steps (Integrated Gradients)
$\hat{y}$		Normalised predicted RUL
$\mathbf{x}$		Input vector, $\mathbf{x} = (x_1, x_2, \ldots, x_d)$
x_j		j-th feature of input vector
$x^{(i)}$		i-th sample in the dataset
$x_j^{(i)}$		Feature j of sample i
x^{cf}		Counterfactual input
x_j^{ref}		Reference (baseline) value for feature j
x'		Baseline input vector (Integrated Gradients)
$\tilde{x}^{(j)}$		Modified input with feature j permuted or ablated
$\tilde{x}_j$		Modified input with feature j replaced by reference value
z		Grid value used in ICE / ALE evaluation
z_k		Bin boundary at position k (ALE)
D		Distance function (counterfactual optimisation)
F		Full feature set, $F = \{1, \ldots, d\}$
K		Number of bins or grid points (ALE / ICE)
$\mathscr{L}$		Loss function
N		Number of samples in the dataset
N_k		Number of samples in ALE bin k
R_c	cycles	Piecewise RUL cap
S		Coalition / subset of features (SHAP)
T		Sliding window length
t	cycles	Current engine cycle
t_{EOL}	cycles	End-of-life cycle
$v(S)$		Value function for coalition S (SHAP)
w_j		Surrogate model coefficient for feature j (LIME)
$w_j^{(i)}$		Surrogate coefficient for feature j, instance i (LIME)
$y^{(i)}$		True normalised RUL for sample i
y'		Target output for counterfactual generation
Greek symbols		
α		Interpolation parameter along the IG integration path, $\alpha \in [0, 1]$
α_k		k-th quadrature point along the IG integration path

$\gamma(\alpha)$		Linear interpolation path from baseline to input (IG)
$\Delta_{j,k}^{(i)}$		Local prediction difference for sample i in ALE bin k
$\hat{\Delta}_{j,k}$		Average local prediction difference in ALE bin k
λ		Regularisation / trade-off parameter (LIME, counterfactual)
ϕ_j		Feature attribution for feature j; method-specific (see SHAP, IG, FA)
ϕ_0		Baseline prediction, $\phi_0 = \mathbb{E}[f(\mathbf{X})]$
$\pi_x(\cdot)$		Locality kernel weighting function (LIME)
$\Omega(g)$		Complexity penalty on surrogate model (LIME)
Feature Importance Scores		
$\mathrm{FI}_j^{\mathrm{ICE}}$		ICE-derived global importance for sensor j
$\mathrm{FI}_j^{\mathrm{FA}}$		Feature Ablation importance for sensor j
$\mathrm{FI}_j^{\mathrm{SHAP}}$		Mean absolute Shapley value for sensor j
$\mathrm{FI}_j^{\mathrm{LIME}}$		Mean absolute surrogate coefficient for sensor j
$\mathrm{FI}_j^{\mathrm{PFI}}$		Permutation Feature Importance for sensor j
$\mathrm{FI}_j^{\mathrm{ALE}}$		ALE range-based importance for sensor j
$\mathrm{FI}_j^{\mathrm{IG}}$		Mean absolute Integrated Gradients attribution for sensor j
$\mathrm{FI}_j^{\mathrm{CF}}$		Counterfactual-derived importance for sensor j
$\mathrm{FI}_j^{\mathrm{method}}$		Generic per-sensor importance score prior to L1 normalisation
$\mathrm{FI}_j^{\mathrm{method,\,L1}}$		L1-normalised feature importance score for sensor j
Functions and Operators		
$\mathrm{ICE}_j(x^{(i)}, z)$		ICE curve for feature j, instance i, at grid value z
$\mathrm{PDP}_j(z)$		Partial Dependence Plot for feature j at value z
$\mathrm{ALE}_j(z)$		Accumulated Local Effects function for feature j at value z
$\mathrm{RUL}_{\mathrm{raw}}(t)$	cycles	Raw remaining useful life at cycle t
$\mathrm{RUL}(t)$	cycles	Piecewise-capped remaining useful life at cycle t
Abbreviations		
ALE		Accumulated Local Effects
C-MAPSS		Commercial Modular Aero-Propulsion System Simulation
CBAM		Convolutional Block Attention Module
CF		Counterfactual Explanations
CNN		Convolutional Neural Network
DANN		Domain-Adversarial Neural Network
FA		Feature Ablation
GRL		Gradient Reversal Layer
GRU		Gated Recurrent Unit
HPC		High-Pressure Compressor
HPT		High-Pressure Turbine

ICE	Individual Conditional Expectation
IG	Integrated Gradients
LIME	Local Interpretable Model-Agnostic Explanations
LPC	Low-Pressure Compressor
LPT	Low-Pressure Turbine
LSTM	Long Short-Term Memory
MRO	Maintenance, Repair and Overhaul
MSE	Mean Squared Error
PDP	Partial Dependence Plot
PFI	Permutation Feature Importance
PHM	Prognostics and Health Management
RMSE	Root Mean Squared Error
RUL	Remaining Useful Life
SHAP	SHapley Additive exPlanations
TCN	Temporal Convolutional Network
XAI	eXplainable Artificial Intelligence

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1 Introduction

Aircraft engines are among the most safety-critical and economically significant components in modern aviation. A single unplanned engine failure during flight can endanger lives. On the ground, unscheduled maintenance events disrupt airline operations, strand passengers and generate costs that cascade through the network, from replacement parts and hangar time to delayed flights and rebooking. Engine maintenance, repair and overhaul (MRO) accounts for a large share of an airline's total operating expenditure and cost difference between a well-timed maintenance intervention and an emergency shutdown can be orders of magnitude.

For most of aviation's history, maintenance has followed either a reactive or a scheduled approach. Reactive maintenance is unacceptable for flight-critical components. Scheduled maintenance (inspect or replace at fixed intervals regardless of actual condition) is safer but expensive, because components are often serviced long before they need to be. Parts with remaining useful life are discarded, inspection windows are conservatively short and aircraft spend more time in hangars than necessary.

Condition-based and predictive maintenance aim to do better (Jardine et al., 2006). Rather than relying on fixed schedules, these approaches use operational data (sensor readings collected during normal operation, also called Snapshots) to estimate how much useful life a component has left. If the remaining useful life (RUL) of an engine can be predicted with reasonable accuracy, maintenance can be scheduled for when it is actually needed: late enough to extract most of a component's service life, early enough to avoid failure. The potential savings are significant, both in direct maintenance costs and in improved aircraft availability. For safety-critical systems like turbofan engines, the added benefit is that degradation can be tracked continuously rather than assessed at discrete inspection intervals, reducing the chance that a developing fault goes undetected between scheduled checks.

Estimating RUL from sensor data is, at its core, a prediction problem. An engine's condition at any point in time is reflected in the readings from its sensors: temperatures, pressures, speeds, vibration levels. The task is to take a history of these readings and predict how many more operational cycles the engine can sustain before failure. Over the past decade, machine learning models have become the dominant approach to this problem (Li et al., 2018, Zheng et al., 2017). They learn the mapping from sensor histories to RUL directly from run-to-failure data, without requiring explicit physical models of the degradation process. On standard benchmark datasets such as C-MAPSS (Saxena et al., 2008), these data-driven methods now achieve prediction errors that would have been difficult to reach with earlier approaches.

But prediction accuracy alone is not sufficient for adoption in safety-critical applications. A model that predicts an engine has 50 cycles of remaining life is only useful if the engineers and decision-makers who act on that prediction can understand why the model arrived at that number. Which sensor readings drove the prediction? Is the model responding to a genuine degradation signal, or to a statistical artefact in the training data? Would the prediction change if a particular sensor were faulty or miscalibrated? These questions matter because the consequences of a wrong prediction in aviation maintenance are not symmetric, overestimating RUL risks a failure in service, underestimating it wastes resources and availability. Without insight

into the model's reasoning, there is no way to assess whether a specific prediction is trustworthy in a specific operational context.

This is the problem that eXplainable Artificial Intelligence (XAI) addresses. XAI encompasses methods that make the internal workings or outputs of machine learning models more interpretable to humans (Barredo Arrieta et al., 2020). Some methods explain individual predictions, why the model gave *this* answer for *this* input. Others characterise global behaviour, which input features matter most across all predictions and how they influence the output. The methods differ in their assumptions, their computational mechanisms and the kinds of questions they answer. A gradient-based method traces importance through the model's own parameters. A perturbation based method observes how predictions change when inputs are modified. A surrogate-based method fits a simpler, interpretable model to approximate the complex one locally. Each perspective captures something real about the model's behaviour, but none captures everything.

This thesis takes the multi-method approach. Eight feature importance methods are applied to a 1D Convolutional Neural Network (CNN) trained for RUL prediction on the NASA C-MAPSS turbofan engine dataset (Saxena et al., 2008). The methods span the major categories of XAI (gradient-based, perturbation-based, Shapley-value-based, marginal-effect based and counterfactual) and are compared systematically across multiple training configurations. The analysis identifies which sensors the model considers most important, evaluates how consistently different methods agree on that ranking, examines how feature importance is distributed across the temporal dimension of the input and tests whether the model can maintain its predictive performance using only the most important sensors.

The thesis is structured as follows. Chapter 2 reviews related work on data-driven RUL prediction and explainability in prognostics, identifying the gap that this thesis addresses. Chapter **??** introduces the theoretical foundations of the eight feature importance methods used. Chapter 4 describes the dataset, model architecture, preprocessing pipeline and experimental setup. Chapter 5 presents the results, from the verification of the implementations through the comparative analysis, temporal attribution, reduced sensor experiments and the discussion of their implications. Chapter 6 summarises the findings and outlines directions for future work.

2 Related Work

2.1 Data-Driven Remaining Useful Life Prediction

The dataset used throughout this thesis is C-MAPSS (Commercial Modular Aero-Propulsion System Simulation), published by Saxena et al. (2008) for the PHM 2008 data challenge. It contains four subsets of simulated turbofan engine run-to-failure trajectories, FD001 through FD004, each with a different combination of operating conditions and fault modes. Every engine unit records 21 sensor measurements and three operational settings per cycle. Degradation is injected into the high-pressure compressor (HPC), the fan, or both, depending on the subset. 2.1 gives the breakdown.

Subset	Training Engines	Test Engines	Operating Conditions	Fault Modes
FD001	100	100	1 (Sea Level)	1 (HPC Deg)
FD002	260	259	6	1 (HPC Deg)
FD003	100	100	1 (Sea Level)	2 (HPC + Fan Deg)
FD004	248	249	6	2 (HPC + Fan Deg)

Table 2.1: Overview of the C-MAPSS dataset subsets Saxena et al. (2008).

FD001 is the simplest case, one operating condition, one fault mode. FD004, with six operating conditions and two fault modes, is the hardest. The multi-condition subsets (FD002 and FD004) contain roughly two and a half times as many engines, reflecting the additional variability. C-MAPSS has appeared in well over a hundred publications since its release and remains the go-to benchmark, mostly because nothing else offers the same scale with controlled degradation across multiple regimes.

Nearly all C-MAPSS studies adopt a piecewise linear (PwL) labelling scheme for the RUL target. Heimes (2008) showed with a multilayer perceptron that predictions improve when the early, healthy portion of each trajectory is assigned a constant RUL rather than a linearly decreasing one. The reasoning is straightforward: sensors show no measurable degradation in the early cycles, so there is nothing in the data to distinguish an engine at cycle 10 from one at cycle 50. Assigning a constant cap avoids forcing the model to learn distinctions that the sensor readings do not support. Heimes set this cap at 130 cycles. Later work has used values between 115 and 130; this thesis uses 125, which is among the most common choices. The value is heuristic, and it matters, particularly on FD002 and FD004, where trajectory lengths vary more.

The first generation of data-driven RUL methods relied on classical machine learning. Wang et al. (2008) won the PHM 2008 challenge with a similarity-based algorithm. Support vector machines, random forests and ensemble approaches followed. Ramasso and Saxena (2014) reviewed this body of work and noted a recurring dependence on hand-crafted health indicators, features built from domain knowledge about which sensors degrade and how. This is effective, but hard to scale.

Deep learning changed that. Sateesh Babu et al. (2016) were among the first to apply a deep Convolutional Neural Network (CNN) to RUL prediction, treating the problem as a regression over multivariate time-series inputs. Li et al. (2018) extended this direction with a deeper architecture and a systematic sliding-window sample preparation scheme, feeding raw sensor data directly into the network without manual feature engineering. Their model outperformed most prior methods on C-MAPSS and became one of the most cited CNN-based approaches in the prognostics literature. Long Short-Term Memory (LSTM)-based approaches appeared around the same time (Wu et al., 2018, Zheng et al., 2017), exploiting the sequential structure of the degradation data. Hybrid architectures combining CNN feature extraction with LSTM temporal modelling soon followed (e.g., Zhao et al. (2017)), and attention mechanisms were layered on top, for instance, Li et al. (2022) added a convolutional block attention module (CBAM) to a CNN-LSTM on C-MAPSS, allowing the model to weight sensor channels and time steps selectively. Transformer-based models and domain adaptation techniques are the most recent additions, through the core task, predicting a scalar RUL from a sensor window, has stayed the same throughout.

The model in this thesis is a 1D-CNN: convolutional layers for feature extraction, fully connected layers for regression. No recurrence, no attention, no gating. That structural simplicity matters here, because the focus is on interpretability. Fewer internal mechanisms mean fewer confounding factors when attributing prediction back to individual input features.

2.2 Explainable Artificial Intelligence in Prognostics

Strong RUL predictions are not enough on their own. In aviation maintenance, where sensor readings feed into decisions about grounding aircraft or scheduling inspections, engineers need to understand what a model is responding to. A prediction driven by physically meaningful degradation signals is useful; one driven by spurious correlations in the training data is dangerous. This tension between predictive accuracy and interpretability runs through the broader machine learning literature Rudin (2019), but it is especially acute in safety-critical prognostics, where a wrong explanation attached to a correct prediction can mislead maintenance decisions just as badly as an incorrect prediction.

Applying XAI to prognostic models is still uncommon. Serradilla et al. (2020) applied Local Interpretable Model-agnostic Explanations (LIME; Ribeiro et al. (2016)) permutation-based feature importance to a random forest-based RUL estimator for industrial bushings, using the explanations alongside domain knowledge to guide model development. Their work demonstrated the principle, that XAI tools can do more than just post-hoc reporting, but was limited to a relatively simple model on non-aviation data.

On C-MAPSS specifically, Hong et al. (2020) were among the first, using SHapley Additive exPlanations (SHAP; (Lundberg and Lee, 2017)) on a deep CNN-Bidirectional LSTM. They first reduced the sensor set through correlation analysis and regularisation, then ran SHAP on the reduced inputs. The resulting force plots broke down individual predictions by sensor contribution. Useful as a proof of concept, but the analysis only covered one explanation method,

and because sensors had already been filtered before the SHAP step, the attributions described a pre-selected feature space rather than what the model would have learned from the full suite.

Baptista et al. (2022) compared SHAP attributions against traditional prognostic metrics, monotonicity, trendability, prognostability, that are commonly used to assess sensor quality before modelling. The two did not always agree: sensor scoring well on prognostic metrics were not necessarily the ones the trained model weighted most heavily. That disconnect says something about the gap between a priori feature assessment and what a network actually extracts from the data during training.

Baptista et al. (2024) then turned to LIME (Ribeiro et al., 2016), applying it to a Gated Recurrent Unit (GRU) Network on C-MAPSS. LIME recovered both global and local feature importance, but the local explanation quality was uneven. Late in the degradation trajectory, where the signal is strong and clear, surrogate fidelity was reasonable. Early in life, with weak degradation signatures and noisy readings, it dropped off sharply, a known limitation of perturbation-based methods on time-series data, but worth confirming empirically.

Alomari et al. (2023) took SHAP in a different direction, using it not just for post-hoc explanation but as part of the feature selection pipeline before training. Similar workflows have since appeared, coupling SHAP-based selection with CNN-LSTM or Temporal Convolutional Networks (TCN)-LSTM architectures. Gawde et al. (2024) applied four methods - SHAP, LIME, Partial Dependence Plots (PDP), and Individual Conditional Expectation (ICE; Goldstein et al. (2015)) - to rotating machinery health monitoring, demonstrating a multi-method comparison, through on a different domain and with classical ML models rather than deep networks.

The methodological range across the C-MAPSS prognostics literature remains narrow. SHAP dominates. LIME appears occasionally. Gradient-based attribution methods, marginal-effect methods, and counterfactual approaches, all well established in the broader XAI field (see **??**), are absent from C-MAPSS RUL work almost entirely. The few multi-method studies that do exist, like Gawde et al. (2024), apply to different equipment and simpler models. On deep networks trained for turbofan prognostics, no comparative evaluation has been carried out.

2.3 Research Gap and Thesis Positioning

The literature reviewed above reveals two things. First, deep learning models for RUL prediction on C-MAPSS have matured rapidly, from basic CNNs and LSTMs to attention-augmented hybrids and Transformers, but the question of what these models actually learn from the sensor data has received far less attention. Second, the small body of work that does address interpretability in this domain relies almost exclusively on SHAP, occasionally LIME, applied in isolation to a single architecture.

This is a problem, because different explanation methods operate on different principles and can produce different, sometimes contradictory, results for the same model and input. A gradient-based method like Integrated Gradients (Sundararajan et al., 2017) traces attributions through the network's own parameters. A perturbation-based method like Permutation Feature Impor-

tance (Breiman, 2001) observes how prediction changes when inputs are shuffled. A marginal-effect method like Accumulated Local Effects (ALE; Apley and Zhu (2020)) characterises the prediction surface shifts as a single feature varies. Relying on one method alone risks mistaking that method's particular perspective for a complete picture of the model's behaviour.

This thesis addresses the gap by applying eight feature importance methods (see **??**) to a 1D CNN trained on a variety of subset combinations of all four C-MAPSS subsets. The contribution is not just an expanded set of explanations, but a systematic comparison, identifying where methods agree on which sensors drive the model's prediction, where they diverge and what effect the different subset combinations have on the model.

3 Fundamentals

This chapter introduces the eight feature importance methods used in this thesis. The methods are divided into model-agnostic approaches, which work with any model, and model-specific approaches, which require access to the model's internal computations. Practical implementation details are covered in Chapter 4.

3.1 Model-Agnostic Methods

Model-agnostic methods treat the model as a black box. They only require the ability to feed inputs into the model and observe the resulting predictions. Let

$$f : \mathbb{R}^d \to \mathbb{R} \tag{3.1}$$

be a trained model mapping an input vector $\mathbf{x} = (x_1, x_2, \ldots, x_d)$ to a scalar output such as a predicted RUL. This definition of f and $\mathbf{x}$ is used throughout this section. By modifying inputs in various ways — sweeping a feature across its range, replacing it with a neutral value, shuffling it across samples — these methods measure how sensitive the model's output is to each input feature.

3.1.1 Individual Conditional Expectation

Individual Conditional Expectation (ICE) shows how a model's prediction changes as one feature is varied, while all other features are held fixed at their observed values (Goldstein et al., 2015). For a sample $x^{(i)}$, the ICE curve for feature j is obtained by substituting a grid of values z into position j:

$$\mathrm{ICE}_j\left(x^{(i)}, z\right) = f\left(x_1^{(i)}, \ldots, x_{j-1}^{(i)}, z, x_{j+1}^{(i)}, \ldots, x_d^{(i)}\right) \tag{3.2}$$

This produces one prediction curve per instance, tracing how the output responds to feature j in isolation. Averaging across all N samples gives the Partial Dependence Plot (PDP):

$$\mathrm{PDP}_j(z) = \frac{1}{N} \sum_{i=1}^{N} \mathrm{ICE}_j\left(x^{(i)}, z\right) \tag{3.3}$$

The PDP describes the average effect of feature j across the dataset. Where individual ICE curves diverge substantially from the PDP, this indicates that the feature's effect depends on the values of other features — a sign of interaction effects that averaging would obscure.

A scalar importance score is derived by computing the mean variance of ICE curves across all instances:

$$\mathrm{FI}_j^{\mathrm{ICE}} = \frac{1}{N} \sum_{i=1}^{N} \mathrm{Var}_z\left[\mathrm{ICE}_j\left(x^{(i)}, z\right)\right] \tag{3.4}$$

This measures how much each instance's prediction varies as feature j is swept across its range. A feature that consistently moves predictions will receive a high score; one the model ignores will not. The resulting score is referred to as $\mathrm{FI}_j^{\mathrm{ICE}}$ throughout this thesis.

3.1.2 Feature Ablation

Feature Ablation measures importance by replacing a feature with a neutral reference value and observing how much the prediction changes (Zeiler and Fergus, 2014). The idea is simple: if removing a feature's information causes a large shift in the output, that feature matters.

For feature j, a modified input $\tilde{x}_j$ is constructed by replacing x_j with a reference value x_j^{ref} — typically the feature mean — while all other features remain unchanged. The local attribution is:

$$\phi_j(\mathbf{x}) = f(\mathbf{x}) - f(\tilde{x}_j) \tag{3.5}$$

Averaging the absolute change across the dataset gives the global importance score:

$$\text{FI}_j^{\text{FA}} = \frac{1}{N} \sum_{i=1}^{N} \left| \phi_j\left(x^{(i)}\right) \right| \tag{3.6}$$

In multivariate time-series data, entire sensor channels can be ablated jointly rather than individual time steps, which is the approach taken in this thesis. One limitation is that results depend on the choice of reference value, and replacing features independently can produce input combinations that are unlikely under the true data distribution when features are correlated (Molnar, 2025). This global score is referred to as FI_j^{FA} throughout this thesis.

3.1.3 Shapley Additive Explanations

SHAP (SHapley Additive exPlanations) attributes a prediction to its input features using ideas from cooperative game theory (Lundberg and Lee, 2017, Shapley, 1953). Each feature is treated as a player in a game, and its attribution is its average contribution to the prediction across all possible subsets of features.

A value function $v(S)$ quantifies the model output when only the features in subset $S \subseteq F = \{1, \ldots, d\}$ are active, with absent features replaced by a reference value. The Shapley value for feature j is:

$$\phi_j = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!\,(d - |S| - 1)!}{d!} \left(v(S \cup \{j\}) - v(S) \right) \tag{3.7}$$

The factorial weight averages the marginal contribution $v(S \cup \{j\}) - v(S)$ over all possible coalition sizes and orderings (Shapley, 1953). The attributions form an additive decomposition of the prediction:

$$f(\mathbf{x}) = \phi_0 + \sum_{j=1}^{d} \phi_j \tag{3.8}$$

where $\phi_0 = \mathbb{E}[f(\mathbf{X})]$ is the expected model output under a background distribution (Lundberg and Lee, 2017). Four axioms uniquely characterise Shapley values: efficiency (attributions sum to the explained quantity), symmetry (identical contributors receive equal attribution), the dummy property (non-contributing features receive zero), and linearity (attributions combine linearly for linear combinations of models).

Exact computation requires evaluating all 2^d possible coalitions, which becomes expensive for large d. In practice, Shapley values are approximated via sampling or weighted regression (Lundberg and Lee, 2017, Štrumbelj and Kononenko, 2014). When features are correlated, the choice of how to handle absent features affects the resulting attributions (Aas et al., 2021). For the purposes of cross-method comparison, the global feature importance score is defined as the mean absolute Shapley value across all instances, $\mathrm{FI}_j^{\mathrm{SHAP}} = \frac{1}{N}\sum_{i=1}^{N} |\phi_j(x^{(i)})|$.

3.1.4 Local Interpretable Model-Agnostic Explanations

LIME explains an individual prediction by fitting a simple linear model to approximate the original model in the neighbourhood of a specific input (Ribeiro et al., 2016). Even a complex model often behaves approximately linearly in a small enough region, and LIME exploits this.

Given an instance $\mathbf{x}$, LIME draws perturbed inputs $\{z_k\}_{k=1}^{K}$ near $\mathbf{x}$ and evaluates each with the model. A locality kernel $\pi_x(z_k)$ down-weights samples that are far from $\mathbf{x}$. A sparse linear model g is then fitted by solving:

$$\arg\min_{g \in \mathscr{G}} \mathscr{L}(f, g, \pi_x) + \Omega(g) \tag{3.9}$$

where $\mathscr{L}$ measures how closely g approximates f under the locality weighting and $\Omega(g)$ penalises complexity (Ribeiro et al., 2016). For a linear surrogate:

$$g(z) = w_0 + \sum_{j=1}^{d} w_j z_j \tag{3.10}$$

the fitted coefficients w_j are the local feature importance scores. Because LIME draws perturbed samples stochastically, explanations can vary slightly across runs. For the purposes of cross-method comparison, the global feature importance score is defined as the mean absolute surrogate coefficient across all instances, $\mathrm{FI}_j^{\mathrm{LIME}} = \frac{1}{N}\sum_{i=1}^{N} |w_j^{(i)}|$.

3.1.5 Counterfactual Explanations

Counterfactual explanations take a different perspective from the attribution-based methods above. Rather than scoring features by their contribution to the current prediction, they ask: what is the smallest change to the input that would shift the model's prediction to a desired target? Features that need to change most are considered the most influential (Wachter et al., 2018).

For an instance $\mathbf{x}$ and a target output y', a counterfactual x^{cf} is found by solving:

$$x^{\mathrm{cf}} = \arg\min_{z} \lambda \cdot \mathscr{L}(f(z), y') + D(z, \mathbf{x}) \tag{3.11}$$

where $\mathscr{L}$ penalises deviation from the target, D is a distance measure (typically L1 or L2) that penalises large changes to the input, and $\lambda > 0$ controls the trade-off between the two

(Wachter et al., 2018). Additional constraints can be added to ensure that the counterfactual remains plausible (Verma et al., 2022) or respects causal structure (Kusner et al., 2018). The optimisation is generally non-convex and multiple valid solutions may exist. The resulting per-sensor magnitude of change, aggregated across instances, is referred to as $\mathrm{FI}_j^{\mathrm{CF}}$ throughout this thesis.

3.1.6 Permutation Feature Importance

Permutation Feature Importance (PFI) measures how much predictive performance drops when the link between a feature and the target is broken by randomly shuffling that feature's values across samples (Breiman, 2001, Fisher et al., 2019). Shuffling destroys any information the feature carries about the target while leaving its marginal distribution unchanged — so any resulting drop in performance can be attributed to the loss of that feature's signal.

For feature j, a permuted dataset $\tilde{x}^{(j)}$ is constructed by randomly shuffling x_j across all samples while all other features remain unchanged. The importance is:

$$\mathrm{FI}_j^{\mathrm{PFI}} = \mathscr{L}\left(f\left(\tilde{x}^{(j)}\right)\right) - \mathscr{L}(f(\mathbf{x})) \tag{3.12}$$

where $\mathscr{L}$ is a performance metric such as mean squared error. A larger value means the model relied more heavily on feature j. One known limitation is that when features are correlated, permuting one independently of the others can generate unrealistic input combinations, which may inflate or distort the resulting estimates (Fisher et al., 2019). This score is referred to as $\mathrm{FI}_j^{\mathrm{PFI}}$ throughout this thesis.

3.1.7 Accumulated Local Effects

Accumulated Local Effects (ALE) estimates the average effect of a feature on model predictions in a way that accounts for correlations between features (Apley and Zhu, 2020). The standard alternative, PDP, averages predictions over the marginal distribution of all features, which can produce evaluations at input combinations that are unrealistic when features are dependent. ALE avoids this by computing prediction differences within narrow local intervals, using only samples that actually fall in that region.

The domain of feature x_j is divided into K intervals at quantile-based grid points $z_0 < z_1 < \cdots < z_K$. For each sample $x^{(i)}$ falling in interval $[z_{k-1}, z_k]$, the local prediction difference is:

$$\Delta_{j,k}^{(i)} = f\left(x_1^{(i)}, \ldots, z_k, \ldots, x_d^{(i)}\right) - f\left(x_1^{(i)}, \ldots, z_{k-1}, \ldots, x_d^{(i)}\right) \tag{3.13}$$

The average within bin k is:

$$\hat{\Delta}_{j,k} = \frac{1}{N_k} \sum_{i: x_j^{(i)} \in [z_{k-1}, z_k]} \Delta_{j,k}^{(i)} \tag{3.14}$$

and these local averages are accumulated to give the ALE function at point z:

$$\mathrm{ALE}_j(z) = \sum_{k: z_k \leq z} \hat{\Delta}_{j,k} \tag{3.15}$$

The function is centred so that:

$$\int \mathrm{ALE}_j(z)\,\mathrm{d}P_{X_j}(z) = 0 \tag{3.16}$$

meaning it reflects deviations from the average prediction rather than absolute values. A scalar importance score is derived as the range of the centred ALE curve,

$$\mathrm{FI}_j^{\mathrm{ALE}} = \max_z \mathrm{ALE}_j(z) - \min_z \mathrm{ALE}_j(z) \tag{3.17}$$

reflecting the total span of predicted RUL variation attributable to that feature. Instability in sparsely populated regions of the feature space remains a limitation.

3.2 Model-Specific Methods

Model-specific methods use quantities internal to the model — gradients, activations, or weights — and are therefore tied to a particular model class. They require access to the model's internals, which restricts their use to differentiable architectures such as neural networks. Within that scope, they are generally more computationally efficient than model-agnostic methods.

3.2.1 Integrated Gradients

Integrated Gradients (IG) is a gradient-based attribution method for differentiable models (Sundararajan et al., 2017). A straightforward approach would be to evaluate the gradient at the input and use it as an importance score. The problem is that gradients only reflect local behaviour and can be uninformative when the model's output saturates. IG addresses this by accumulating gradients along a straight-line path from a reference baseline to the input being explained, building a more complete picture of each feature's contribution.

Let $\mathbf{x}$ be the input and x' a baseline representing an uninformative reference state, for example an all-zeros vector. The path between them is:

$$\gamma(\alpha) = x' + \alpha(\mathbf{x} - x'), \quad \alpha \in [0,1] \tag{3.18}$$

The attribution for feature j is the path integral of the partial derivative of f with respect to x_j, scaled by the displacement between baseline and input:

$$\phi_j(\mathbf{x}) = (x_j - x'_j) \int_0^1 \frac{\partial f(\gamma(\alpha))}{\partial x_j}\,\mathrm{d}\alpha \tag{3.19}$$

In practice, the integral is approximated using a Riemann sum with m steps:

$$\phi_j(\mathbf{x}) \approx (x_j - x'_j)\,\frac{1}{m}\sum_{k=1}^{m} \frac{\partial f(\gamma(\alpha_k))}{\partial x_j}, \qquad \alpha_k = \frac{k}{m} \tag{3.20}$$

IG satisfies the completeness property — attributions sum exactly to the difference in model output between input and baseline:

$$\sum_{j=1}^{d} \phi_j(\mathbf{x}) = f(\mathbf{x}) - f(x') \tag{3.21}$$

Two further properties hold: sensitivity (a feature responsible for a prediction difference always receives non-zero attribution) and implementation invariance (functionally equivalent models produce identical attributions) (Sundararajan et al., 2017). The choice of baseline affects the resulting attributions and no universally correct choice exists. The global feature importance score is defined as the mean absolute attribution across all instances, $\mathrm{FI}_j^{\mathrm{IG}} = \frac{1}{N}\sum_{i=1}^{N} |\phi_j(x^{(i)})|$, and is referred to as $\mathrm{FI}_j^{\mathrm{IG}}$ throughout this thesis.

4 Methodology

This chapter describes the experimental setup used to analyse feature importance in neural network-based RUL prediction. It covers the dataset and preprocessing pipeline, the model architecture and its three configurations, the training procedure, and the implementation of feature importance methods. The models themselves are not the primary contribution of this thesis; they serve as controlled predictive systems under which feature importance behaviour can be systematically studied across different input configurations.

4.1 Dataset and Data Preparation

4.1.1 Features set / sensors used

The raw input space of each CMAPSS sub-dataset consists of three operational settings and 21 sensor measurements per time step. Eight sensors: s1, s5, s6, s10, s14, s16, s18, and s19, were identified as non-informative due to near-constant values across all trajectories and were excluded from the analysis. The remaining 13 sensor channels, listed in 4.1, form the feature set used throughout this work.

Table 4.1: Sensor features retained after non-informative sensor removal.

Sensor ID	Physical Description
s2	LPC outlet temperature
s3	HPC outlet temperature
s4	LPT outlet temperature
s7	HPC outlet pressure
s8	Fan speed
s9	Core speed
s11	HPC outlet static pressure
s12	Ratio to Fuel flow to Ps30
s13	Corrected fan speed
s15	Bypass ratio
s17	Bleed enthalpy
s20	HPT cool air flow
s21	LPT cool air flow

Each sensor channel is scaled independently to the range [0, 1] using a MinMax scaler. Because sensor readings vary systematically with operating condition, scaling is applied separately within each of the distinct operating regimes present in the data. The scaler is fitted exclusively on the training set.

Input sequences are constructed using a sliding window of length $T = 70$ time steps and stride 1, yielding tensors of shape (N x 13 x 70), where N is the number of sequences. The label

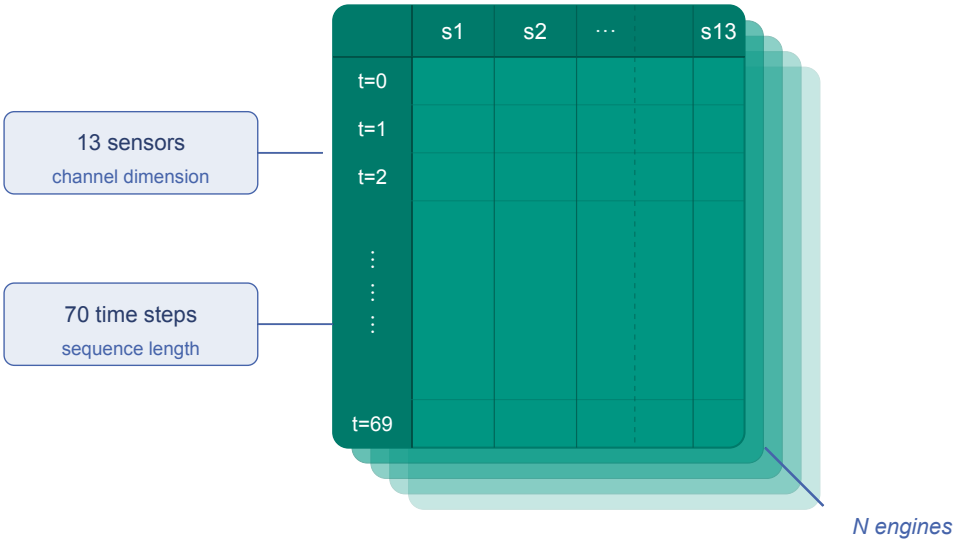

Figure 4.1: Structure of the input tensor for a single sliding window sample, consisting of 13 sensor channels over 70 consecutive time steps. Each slice in the depth direction represents one engine unit.

associated with each window is the RUL value at the final time step of that window. The resulting input structure is illustrated in Figure 4.1.

4.1.2 Target definition (RUL label)

The raw RUL for each engine trajectory is computed as the difference between the end-of-life cycle t_{EOL} and the current cycle t:

$$\text{RUL}_{\text{raw}}(t) = t_{\text{EOL}} - t \tag{4.1}$$

A purely linear RUL label is known to dominate model training with the healthy regime, where no meaningful degradation signal is present. To address this, a piecewise linear transformation is applied using a maximum RUL threshold $R_c = 125$ cycles. Early-life cycles, where the true RUL exceeds R_c, are assigned a constant label of R_c, under the assumption that the engine shows no detectable degradation in this regime. Only once the trajectory enters the degradation region does the label decrease linearly toward zero at failure:

$$\text{RUL}(t) = min(\text{RUL}_{\text{raw}}(t), R_c). \tag{4.2}$$

This transformation prevents the model from being overwhelmed by healthy-state samples and focuses learning on the degradation region, which carries the operationally relevant signal.

Following the piecewise transformation, RUL labels are normalized to [-1, 1] using a min-max scaling based on the training set:

$$\hat{y} = 2 \cdot \frac{\mathrm{RUL}(t) - \mathrm{RUL}_{\min}}{\mathrm{RUL}_{\max} - \mathrm{RUL}_{\min}} - 1, \tag{4.3}$$

where $\mathrm{RUL}_{\min}$ and $\mathrm{RUL}_{\max} = R_c = 125$ by construction, so that a healthy engine at the cap maps to +1 and an engine at failure maps to -1. The same scaling parameters are applied to the test set without re-fitting.

4.2 Model Setup

All model variants used in this thesis share a common architecture based on a Domain-Adversarial Neural Network (DANN) . The DANN consists of three components: a convolutional feature extractor, a label predictor, and an optional domain classifier connected via a Gradient Reversal Layer (GRL). Since the primary purpose of the models here is to serve as a stable substrate for feature importance analysis rather than to maximise transfer performance, the domain classifier branch is used selectively and the architecture is held fixed across all configurations.

The **Feature Extractor** processes the input tensor of shape $(Nx13xT)$ through four successive one-dimensional convolutional layers. The number of filters follows a halving schedule: 128, 64, 32, and 16 channels respectively, each with kernel size 5 and same-length padding of 2. After each convolutional layer, a LeakyReLU activation with negative slope 0.1 is applied. The temporal dimensions is collapsed using a global average pooling operation (AdaptiveAvgPool1d), which produces a fixed-length representation of 16 values regardless of input sequence length. A final fully connected layer projects this to a feature vector of length 64. Note that the slab height in Figure 4.2 is a visual convention representing the reducing channel count and does not imply a reduction in the temporal dimension, which remains 70 throughout.

The **Label Predictor** maps the 64-dimensioinal feature vector to a scalar RUL estimate through two fully connected layers: Linear $(64 \to 32)$ and Linear $(32 \to 1)$, each preceded by a ReLU activation. The output is a single value representing the predicted RUL.

4.3 Training Procedure

All models are trained to minimise the mean squared error (MSE) between the predicted and true normalised RUL values:

$$\mathscr{L} = \frac{1}{N} \sum_{i=1}^{N} (\hat{y}^{(i)} - y^{(i)})^2. \tag{4.4}$$

The models are trained with the following subset combinations.

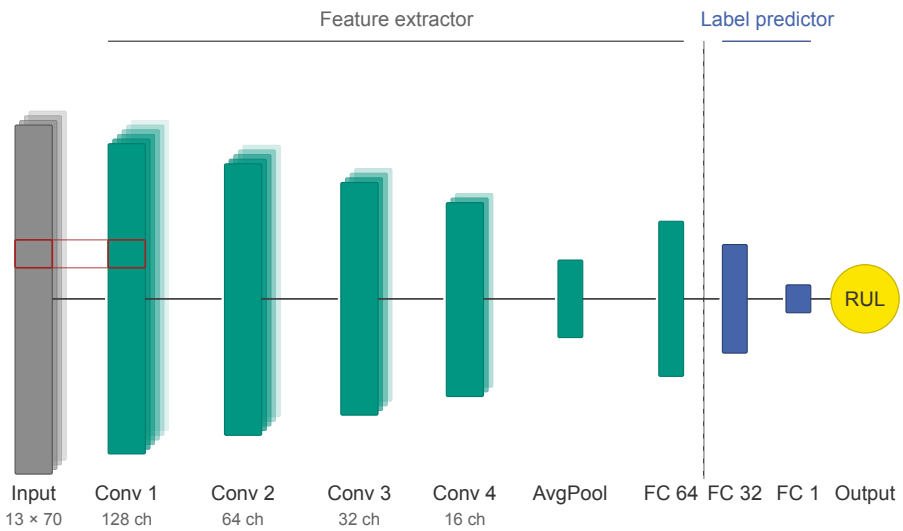

Figure 4.2: Architecture of the 1D convolutional neural network. The red window illustrates the receptive field of the convolutional kernel ($k = 5$). Green: feature extractor, blue: label predictor.

4.3.1 Model Configurations

Three training configurations are studied, differing only in the subset of CMAPSS data used during training. The four single-dataset models each train on one sub-dataset in isolation and serve as the baseline for feature importance behaviour under a single operating condition. The

six pairwise models each train on the concatenated data from two sub-datasets, allowing examination of how feature importance shifts when a model must generalise across two distinct operating regimes simultaneously. The full-dataset model trains on all four sub-datasets combined and represents the most demanding generalisation scenario. Table 4.2 gives an overview of all eleven model configurations.

Table 4.2: Overview of all model configurations used in this work.

Configuration	**Training data**	**No. of models**
Single	FD001	
	FD002	
	FD003	
	FD004	4
Pairwise	FD001 + FD002	
	FD001 + FD003	
	FD001 + FD004	
	FD002 + FD003	
	FD002 + FD004	
	FD003 + FD004	6
Full	FD001 + FD002 + FD003 + FD004	1
Total		**11**

4.4 Implementation of Feature Importance Methods

All eight feature importance methods are applied post-hoc to the trained models, without any modification of the learned weights. A central implementation challenge shared across all methods is the multivariate time-series structure of the input: each sample is a tensor of shape (N x 13 x 70), where the 13 sensor channels and 70 time steps together constitute the model's input space. Most standard feature importance implementations assume a tabular input where each column corresponds to an independent scalar feature. Applying these methods to a sequence model therefore requires deliberate design decisions about how features are defined, how perturbations are applied and how attributions over the temporal dimension are reduced to a per-sensor importance score. Unless stated otherwise, the full temporal extent of a sensor channel is treated as a single unit, perturbations are applied uniformly across all 70 time steps of a given sensor simultaneously and the resulting attributions are aggregated over time to yield a scalar importance score per sensor. This choice reflects the physically motivated view that a sensor's contribution to the RUL prediction should be attributed to the sensor as a whole rather than decomposed into individual time steps, which would result in a 910-dimensional attribution space that is difficult to interpret and compare across methods.

Individual Conditional Expectation (ICE): For each of the 13 sensors, a uniform grid of 20 values is constructed between the 5th and 95th empirical percentile of that sensor's observed values across the dataset. The percentile bound are used in place of the global minimum and maximum to avoid evaluating the model in the tails of the distribution, where training data is sparse and prediction may be unreliable. For each grid value g and each instance in the dataset, the entire time series of sensor j is replaced by the constant value g, that is, $x_{j,t} = g$ for all $t \in [1, 70]$, while all remaining sensor channels are kept at their observed values. This produces one ICE curve per instance, tracing how the predicted RUL changes as sensor j is swept across its range in isolation. The PDP is then computed as the pointwise mean of all ICE curves. Feature importance is derived as ICE-FI$_j$ (see **??**), measuring the mean variance of ICE curves across all instances.

Feature Ablation: The implementation follows the grouped ablation scheme provided by Captum's *FeatureAblation*, in which all 70 timesteps belonging to a given sensor channel are masked simultaneously as a single group rather than independently. This sensor-grouped formulation is essential for the present setting: ablating individual timesteps in isolation would produce attributions across 910 independent input positions, making cross-method comparison difficult and reducing the physical interpretability of the results. When sensor j is ablated, its entire time series is replaced by the training set mean computed independently for each channel, representing the expected signal in the absence of informative variation. Attributions are defined as the absolute change in prediction relative to the unperturbed output, and global feature importance is obtained by averaging these absolute differences over all instances in the evaluation set.

Shapley Additive Explanations: To maintain consistency with the grouped ablation scheme described above, the 13 sensor channels are treated as 13 players in the cooperative game rather than operating on the 910-dimensional flattened input. Exact Shapley values are computed by enumerating all $2^{13} = 8192$ possible sensor coalitions and evaluating the model on each, which is computationally tractable for this group size. When a sensor group is absent from a coalition, its timesteps are replaced by a mean valued baseline as in the ablation method. For each coalition S, the masked input is constructed by replacing the features of all sensors not in S with the baseline and the model is queried to obtain $v(S)$. The Shapley values are then computed from these cached coalition outputs, yielding a per-sensor attribution for each instance in the dataset. A visualisation of the Shapley values is provided in the results chapter and implemented using the *shap* library.

Local Interpretable Model-Agnostic Explanations: The implementation uses Captum's `LIME` with a Lasso linear surrogate as the interpretable model class. Since LIME operates by perturbing the input in an interpretable feature space and fitting a weighted linear regression, the (13 x 70) input tensor is first flattened to a 910-dimensional vector. For each instance to be explained, 4000 perturbed samples are drawn in the neighbourhood of the input, each evaluated by the model. A proximity-weighted Lasso regression($\alpha = 0.01$) is then fitted to approximate the model locally, with the regularisation term enforcing sparsity in the resulting attribution vector. The zero vector is used as the reference baseline. After fitting, the 910-dimensional weight vector is reshaped back to (13 x 70) and the time dimension is averaged to obtain sensor-level

importance scores. The Lasso regularisation parameter α was selected to balance sparsity with attribution stability across instances.

Counterfactual Explanations: Counterfactual explanations are generated using the `DiCE` library, which implements a gradient -based search for minimally modified inputs that shift the model's prediction into a specified target range. As `DiCE` requires a tabular input representation, the (13 x 70) tensor is flattened to a 910-dimensional feature vector before being passed to the `DiCE` data interface, which is initialised using the training set to define plausible feature ranges. For each query instance, the desired output range is specified as a window around the original prediction, so that the counterfactual represents a plausible alternative trajectory with a meaningfully different predicted RUL. The distance between the original and counterfactual input is penalised using an L_1 norm, which encourages sparse modifications, that is, changes concentrated in a small number of features. Feature importance is subsequently derived by computing the magnitude of change per sensor, aggregated across the 70 timesteps, which identifies which sensors were most frequently or substantially modified to achieve the target prediction shift.

Permutation Feature Importance: For each sensor channel, its values are permuted randomly across all N samples in the evaluation set, while all other channels remain at their original values. This destroys the statistical association between that sensor and the target RUL without altering the marginal distribution of the sensor's values, so any resulting drop in predictive performance can be attributed to the loss of that sensor's information. Performance is measured as MSE on the normalised RUL scale. To account for variability in the permutation, the procedure is repeated with 5 independent random permutations and the mean and standard deviation of the MSE increase are reported per sensor. The use of 5 repeats provides a sufficiently stable estimate for 13 sensors while keeping the total number of model evaluation manageable.

Accumulated Local Effects: The domain of each sensor is partitioned into 20 bins using quantile-based bin edges, so that each bin contains approximately the same number of samples. Quantile-based binning is preferred over uniform binning because sensor distributions may be skewed after operating-condition-specific normalisation and equal-frequency bins ensure that local effect estimates are based on sufficient data throughout the feature range. Bin membership for each sample is determined by the mean value of that sensor's time series, which reduces the (N x L) sensor values to a scalar per sample for assignment purposes. Within each bin, the local prediction difference is computed by replacing the entire time series of sensor j with the lower and upper bin boundary values in turn, and recording the resulting change in model output. These local differences are averaged within each bin and then accumulated across bins following **??**. The accumulated effect curve is centered to zero mean over the empirical distribution of sensor values. A scalar importance score is derived as the range of the centered ALE curve, reflecting the total span of predicted RUL variation attributable to that sensor under the local conditional distribution.

Integrated Gradients: The implementation uses Captum's `IntegratedGradients` with 100 integration steps and the Gauss–Legendre quadrature approximation, which provides a more accurate estimate of the path integral than a simple Riemann sum for the same number of steps. The `multiply_by_inputs` flag is set to `True`, which scales each gradient by

the difference between the input and the baseline. This ensures that the completeness property holds, i.e. the sum of attributions over all input dimensions equals the difference between the model's prediction at the input and at the baseline. A zero-valued tensor of shape $(1 \times 13 \times 70)$ is used as the baseline, consistent with the choice made for SHAP and Feature Ablation. Since all sensor values lie in $[0,1]$ after MinMax normalisation, a zero baseline represents a meaningful reference state corresponding to the minimum observed sensor value in each channel. Attributions are computed per instance over the full (13×70) input space, absolute values are taken to reflect attribution magnitude regardless of sign, and the temporal dimension is averaged to yield a per-sensor importance score of shape $(13,)$.

4.5 Unified Feature Importance Representation

Because the eight feature importance methods produce attributions in different units and on different scales, a unified representation is required before any cross method comparison can be made. For each method, the raw per-sensor score $\mathrm{FI}_j^{\text{method}}$ is normalised using L1 normalisation as shown in Equation 4.5, so that scores sum to one across all sensors for any given method and configuration.:

$$\mathrm{FI}_j^{\text{method, L1}} = \frac{|\mathrm{FI}_j^{\text{method}}|}{\sum_{k=1}^{13} |\mathrm{FI}_k^{\text{method}}|} \tag{4.5}$$

so that the importance scores for all features sum to one for any given instance or aggregate. This normalisation ensures that the magnitude of attributions is comparable across methods regardless of their original scale, enabling direct visual and quantitative comparison in the results chapter.

5 Results

5.1 Verification of FI algorithms

Before applying the eight feature importance methods to the C-MAPSS models, a controlled verification experiment was carried out to confirm that the implementations behave as expected when ground truth is known. The idea is simple: train a model where the true feature importances are unambiguous, then check whether each method recovers them.

The setup uses a linear function of tn input variables:

$$f(x) = 5x_0 + 3x_1 + 2x_2 + 1.5x_3 + 1X_4 + 0x_5 + 0x_6 + 0x_7 + 0x_8 + 0x_9 \tag{5.1}$$

The first five features carry all the predictive information, with known weights of 5, 3, 2, 1.5 and 1. The remaining five features (x_5 through x_9) have zero weight, they are pure noise. A single-layer linear neural network without activation functions was trained on 4000 samples drawn from a uniform distribution over [0, 10]. After training, the learned weights matched the true weights to four decimal places, confirming that the model itself was not a source of error.

All eight feature importance methods were then applied to the trained model using the test set. The resulting attributions were scaled to compare true weights with the trained weights, allowing a direct comparison. 5.1 shows the results.

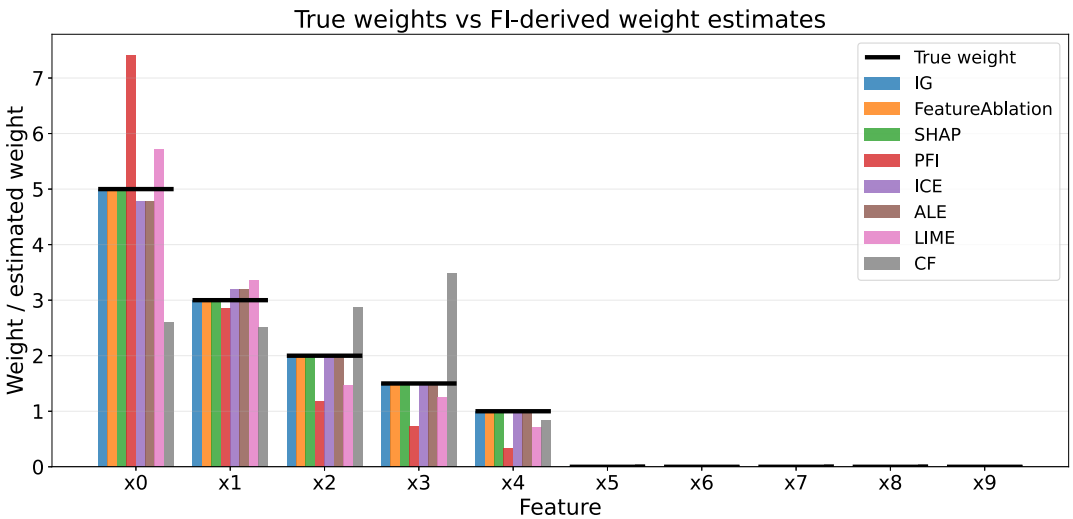

Figure 5.1: Verification of feature importance implementations on a linear model with known weights. Bars show estimated weight; black markers indicate the true weights.

Six of the eight methods, Integrated Gradients, Feature Ablation, SHAP, ALE, ICE and LIME, closely reproduce the true weight distribution. Their bars track the ground truth across all ten

features, correctly assigning near-zero importance to x_5 through x_9 and recovering the correct ranking and approximate magnitudes for x_0 through x_4. Minor deviations exist, ICE slightly overestimates the importance of x_0 relative to the other active features, but these are within the range expected from sampling variability.

Permutation Feature Importance (PFI) and Counterfactual Explanations (CF) show larger deviations, though for different reasons.

PFI overestimates x_0 substantially, roughly 2.5 in estimated weight terms, versus the true value of 5, while compressing the remaining active features downward. The ranking among x_0 through x_4 is preserved and x_5 through x_0 are correctly identified as irrelevant. The skew toward x_0 is a known property of PFI on correlated or high-variance features: because PFI measures the increase in prediction error when a feature is permuted, the feature with the largest absolute weight causes the largest error disruption when shuffled, and this effect scales nonlinearly with the weight magnitude under squared-error metrics like MSE. The relative ordering is correct, but the magnitude distribution does not directly reflect the underlying weight ratios.

Counterfactual Explanations show a different kind of deviation. Rather than preserving the ranking with distorted magnitudes, CF partially misorders the active features, in the raw weieght estimates, x_3 receives the highest estimated weight and the overall distribution across x_0 through x_4 does not follow the true ordering . This is not unexpected. Counterfactual methods search for the smallest input change that shifts the prediction into a target range. That search depends on the local geometry of the prediction surface and the permitted perturbation ranges, not directly on the magnitude of learned weights. In a linear model the prediction surface is uniformly sloped, which removes the kind of local structure that counterfactual search i designed to exploit, making this an inherently unfavourable test case for the method. On more complex models with nonlinear interactions, such as the 1D CNN used on C-MAPSS, counterfactual explanations may behave more meaningfully.

The key outcome of this verification is that all eight methods correctly distinguish relevant feature from irrelevant ones. The six gradient-based, perturbation-based and marginal-effect methods additionally recover the correct ranking and approximate magnitudes. PFI recovers the ranking but distorts magnitudes. CF struggles with both ranking and magnitudes on this linear test case, but still separates the two groups cleanly. These results provide confidence that the implementations are functioning correctly and that observed differences on C-MAPSS reflect genuine properties of the methods rather than implementation errors.

5.2 Individual Method Results

Before comparing the eight methods against each other, it is worth looking at the kind of output they produce individually. This section shows representative ICE curves and SHAP plots, all from the single-FD001 model, the simplest configuration and therefore the easiest to interpret. Complete plots for all model configurations are provided in Appendix A.1

5.2.1 ICE Curves

ICE plots show how the model's RUL prediction changes as a single sensor value is varied across its observed range, with one curve per test instance. The black line is Partial Dependence (PD) curve, the average across all instances. For s9, the individual ICE curves spread across a

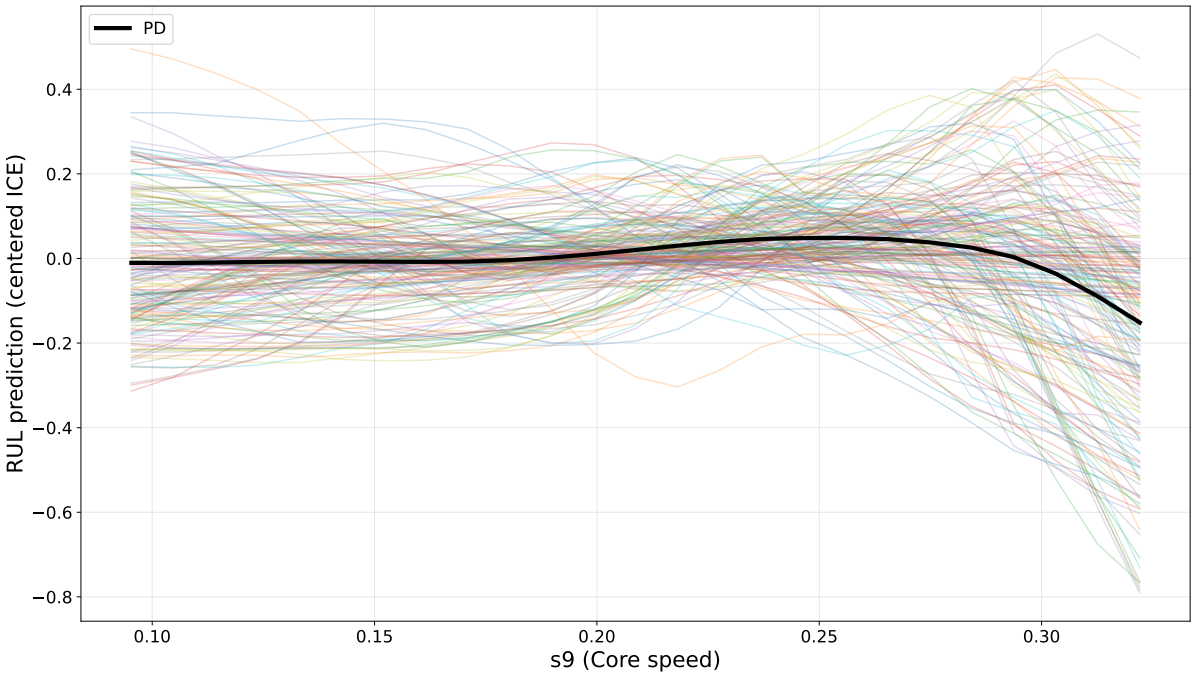

Figure 5.2: Centered ICE curves and PD line (black) for sensor s9 on the single-FD001 model. Each coloured line represents one test instance.

range of roughly ± 0.4 in centered RUL, with several reaching ± 0.6 or beyond at the extremes of the sensor range. The PD curve shows a mild negative trend, higher s9 values are associated with slightly lower predicted RUL, but the more telling aspect is the spread itself. The wide dispersion of individual curves means that the effect of s9 on the prediction depends heavily on the values of the other sensors for that particular engine instance. The model is not using s9 in a simple, unifrom way; its influence interacts with the broader sensor context.

Sensor s3 looks entirely different. The PD curve is effectively flat and the individual ICE curves are tightly clustered within ± 0.1. Varying s3 across its full range barely moves the prediction regardless of context. The model has learned to largely ignore this sensor, consistent with the comparative importance rankings in the next section, where s3 appears near the bottom across all methods.

The contrast between these two sensors illustrates what ICE curves are good at: they do not just say "this sensor matters more than that one," they show *how* a sensor's effect manifests, whether it is roughly uniform across instances or highly variable, suggesting interactions with other features.

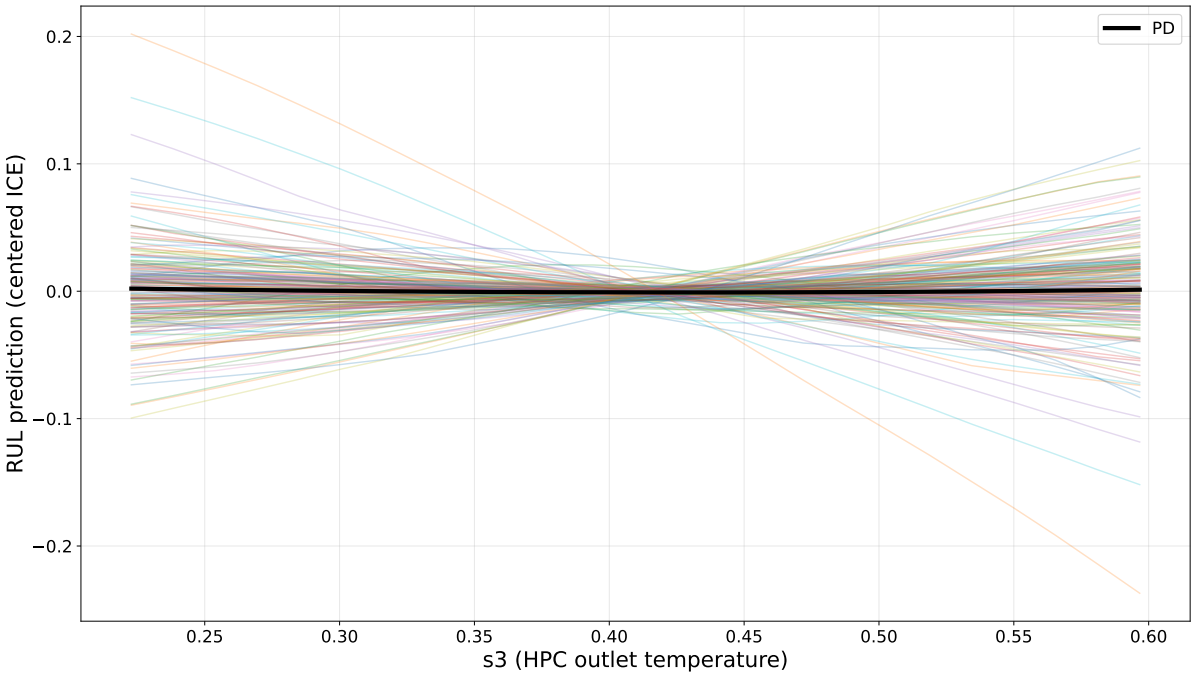

Figure 5.3: Centered ICE curves and PD line (black) for sensor s3 on the single-FD001 model. Each coloured line represents one test instance.

5.2.2 SHAP Analysis

Figure 5.4 shows the SHAP beeswarm plot for the single-FD001 model. Each dot represents one test instance; horizontal position indicates the SHAP value (contribution to the prediction) and colour indicates the sensor reading for that instance (red for high, blue for low). Sensor s9 stands out immediately, it has by far the widest spread of SHAP values, ranging from approximately -0.45 to $+0.40$. No other sensor comes close in magnitude. Sensors s7 and s2 form a second tier, with SHAP values reaching roughly ± 0.2. The colour gradient shows a clear directional pattern for s9: red dots (high sensor values) tend toward negative SHAP values, while blue dots (low values) cluster on the positive side. Sensors s7, s2 and s12 form a second tier, with SHAP values reaching roughly ± 0.15 to ± 0.20. Below that, s4, s11, s15 and s8 all show moderate spread, and the remaining sensors, s20, s21, s13, s17, s3, are tightly clustered around zero. The ranking is more distributed than the bar chart comparisons might suggest: rather than a single dominant sensor and sharp drop-off, the beeswarm reveals a gradual decline in influence across the top seven or eight sensors.

Figure 5.5 shows a SHAP waterfall plot for a single test instance, one engine unit at one point in time, breaking down how the model moves from the expected prediction $E[f(X)] = 0.007$ to the instance-specific prediction $f(x) = -0.738$. This instance sits near the end of the degradation trajectory, where the model predicts a strongly negative normalised RUL. Almost every sensor pushes the prediction downward from the baseline. Sensors s7 (-0.14) and s12 (-0.13) contribute the largest negative shifts. Sensor s9 is the only feature pushing in the opposite direction ($+0.11$), partially counteracting the downward pull. The remaining sensors, s8 (-0.11),

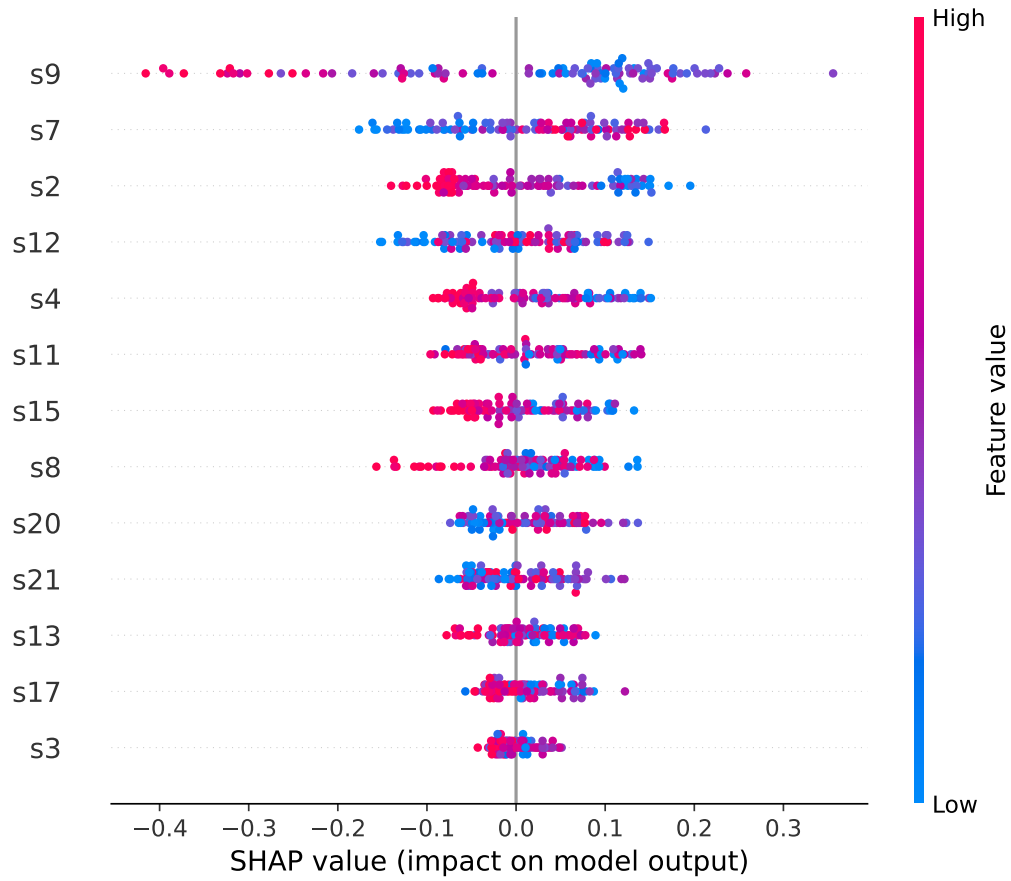

Figure 5.4: SHAP beeswarm plot for the single-FD001 model. Each dot is one test instance; horizontal position shows the SHAP value, colour indicates the sensor reading (red = high, blue = low). Sensors are ordered by decreasing mean absolute SHAP value.

s20, s4, s13, s11 (each around −0.07), s2 (−0.06) and the rest with smaller contributions, all add incrementally to the negative prediction. The waterfall illustrates something the beeswarm alone cannot show as clearly: on a late-life instance, the model draws on nearly every sensor to build up a low RUL prediction, rather than relying on a single dominant signal.

The SHAP plots add directional information that the bar chart comparisons in the next section cannot provide. Where the bar charts show *how much* each sensor matters in aggregate, the beeswarm and waterfall show in *in which direction* and *for which instances*, whether a high sensor reading pushes predicted RUL up or down. The waterfall also makes clear that individual predictions can be driven by a different sensor than the one that ranks highest in aggregate: here, s7 and s12 dominate this particular instance, even though s9 leads in the overall beeswarm ranking. This directional and instance-level detail is useful for sanity-checking the model against physical expectations (see 5.7)

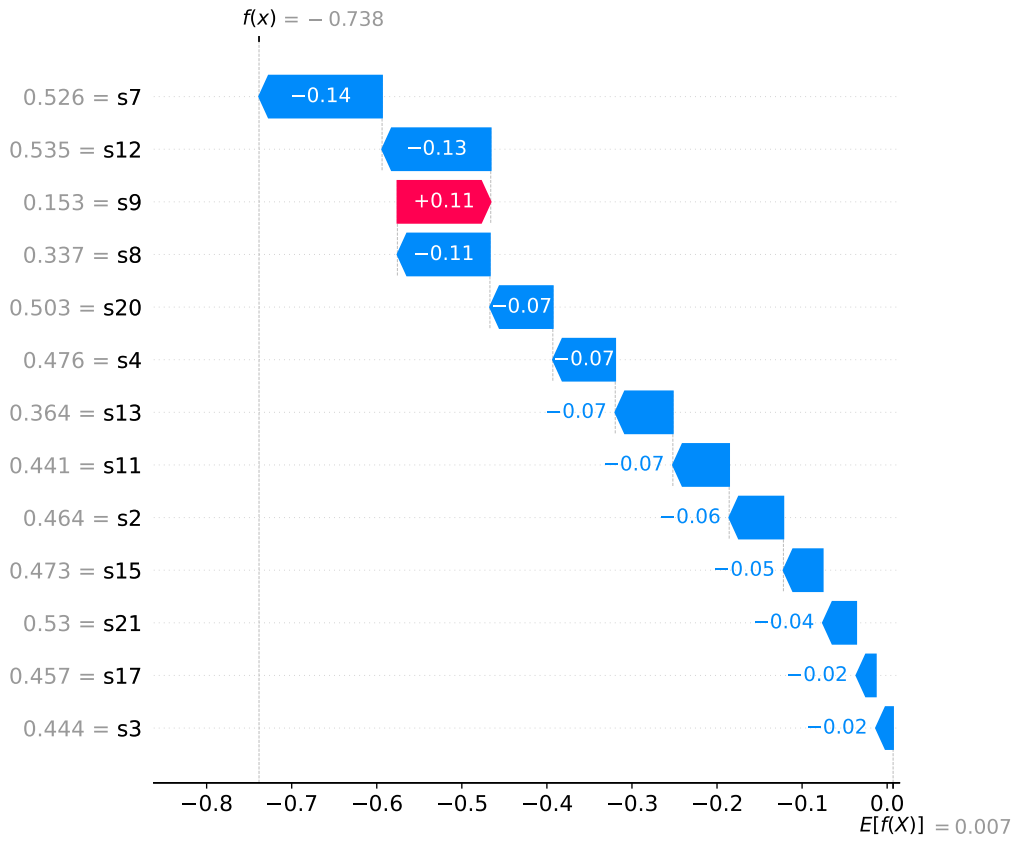

Figure 5.5: SHAP waterfall plot for a single test instance on the single-FD001 model. Each bar shows one sensor's contribution, shifting the prediction from the baseline $E[f(X)] = 0.007$ to the instance output $f(x) = -0.738$.

5.3 Comparative Feature Importance Across Methods

The core of this analysis is the side-by-side comparison of all eight methods on the same model and dataset. Each figure shows L1-normalised sensor importance from every method, with sensors ordered by decreasing average importance. Four representative configurations are shown: models trained on FD001 alone, FD002 alone, the FD001-FD002 pair, and all four subsets combined. Remaining configurations are included in Appendix A.2.

5.3.1 Single-Dataset Models

On the single-FD001 model, s9 is the most important sensor according to most methods. ICE assigns it roughly 0.43, PFI around 0.34, and the gradient- and perturbation-based methods (IG, SHAP, FA) converge around 0.19-0.20. The next tier, s7, s12, and s2, is where methods start to diverge. LIME assigns s7 a normalised importance of roughly 0.34, well above what the other methods give it. ALE places s12 and s4 higher than most alternatives. The broad picture is consistent though: s9 leads, a group of four or five sensors follows with moderate importance and the remainder contributes little.

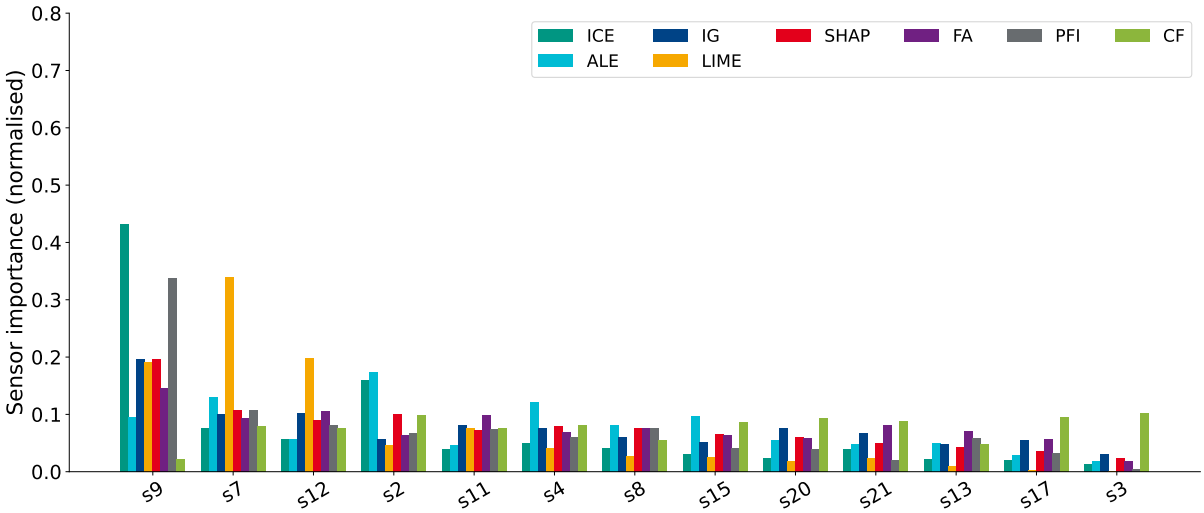

Figure 5.6: L1-normalised sensor importance across all eight methods for the single-FD001 model. Sensors are ordered by decreasing mean importance.

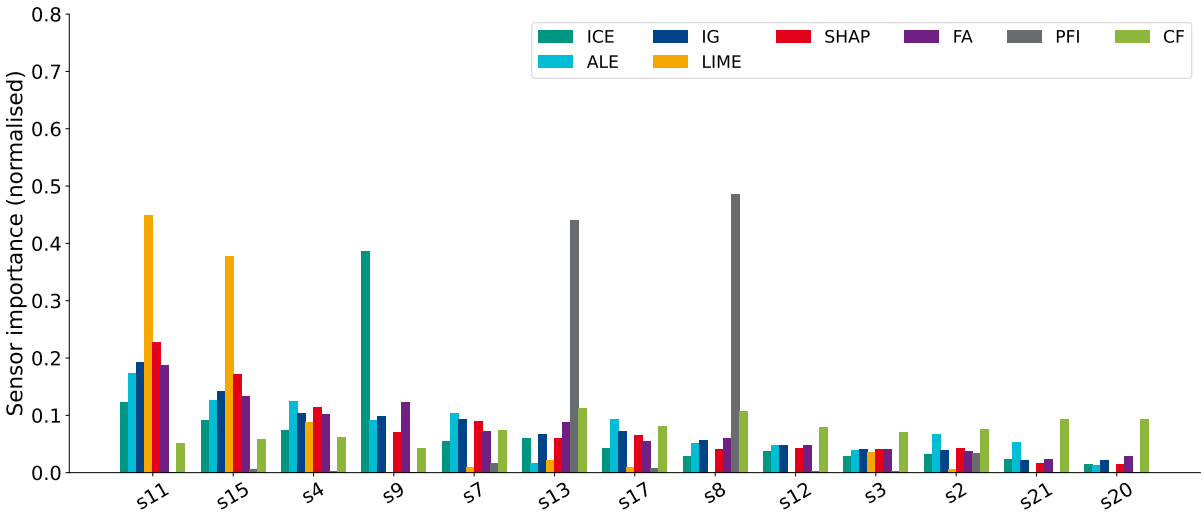

Figure 5.7: L1-normalised sensor importance across all eight methods for the single-FD002 model. Sensors are ordered by decreasing mean importance.

FD002 changes the picture substantially. With six operating conditions instead of one, the model faces more complex input distributions and sensor rankings shift accordingly. Sensor s11 emerges as the most important for IG, ALE, SHAP and FA, the four methods that tended to agree most closely on FD001 as well. LIME concentrates even more aggressively, assigning s11 roughly 0.45 and s15 around 0.38, leaving little importance for anything else. PFI produces extreme outlier spikes, s8 receives nearly 0.49 and s13 around 0.44, not reflected by any other method. These PFI spikes are reminiscent of the magnitude distortion observed in the verification experiment and suggest that PFI's MSE-based scoring amplifies the apparent importance of features whose permutation causes large error in isolated instances.

ICE remains erratic in regards to assigning s9 the highest importance at around 0.39 while other

methods rank it considerably lower.

5.3.2 Multi-Dataset Models

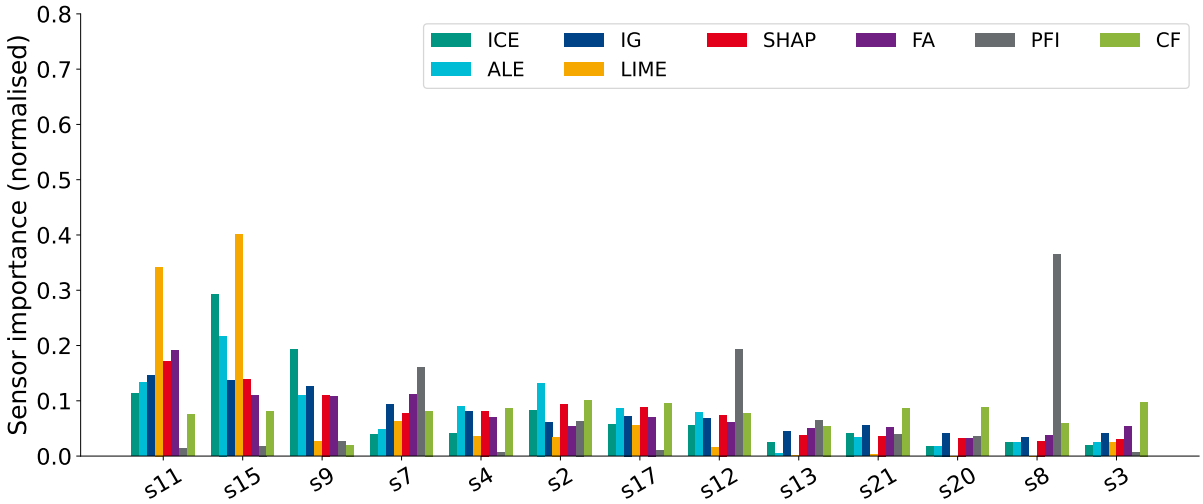

Figure 5.8: L1-normalised sensor importance across all eight methods for the pairwise-FD001-FD002 model. Sensors are ordered by decreasing mean importance.

When the model is trained on FD001 and FD002 together, the rankings stabilise somewhat. Sensors s11 and s15 emerge as the top two across most methods, with s9, s7 and s4 forming a second tier. The pairwise model has seen both single-condition and multi-condition data and the resulting importance reflects a compromise: sensors that dominated under one condition (s9 for FD001, s11 for FD002) both retain importance, but neither dominates as strongly as it did alone. PFI still produces an isolated spike on s8 (~ 0.36), unsupported by any other method. LIME remains heavily concentrated on s11 and s15, allocating around 0.35 and 0.40 respectively.

The all-dataset model shows the most agreement across methods. Sensor s11 is the clear front-runner, IG, SHAP, FA, ALE and ICE all place it first, with normalised importance between 0.10 and 0.19. LIME assigns it roughly 0.48, continuing its pattern of over concentration. Sensors s4 and s9 follow in the second tier, with s12, s2 and s15 in the middle ground. PFI deviates again, spiking on s12 (~ 0.30) and s13 (~ 0.30) while giving s11 minimal importance. CF distributes importance more evenly than in single-dataset cases but still does not match the consensus ranking.

5.3.3 Patterns Across Configuration

Several patterns hold across all four configurations.

The gradient-based and Shapley-based methods, IG, SHAP, and FA tend to cluster together. They rarely disagree on which sensor is most important and their normalised magnitudes are usually within a few percentage points of each other. On any individual model, ALE often

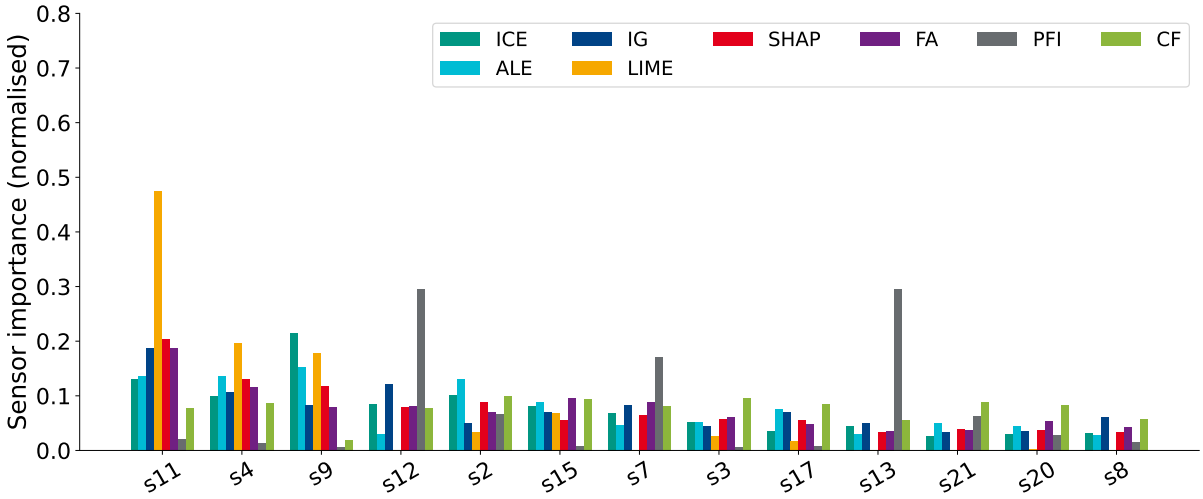

Figure 5.9: L1-normalised sensor importance across all eight methods for the model trained on all subsets. Sensors are ordered by decreasing mean importance.

agrees with this cluster as well, though its cross-configuration stability is lower, a point examined quantitatively below.

LIME consistently over-concentrates importance onto one or two sensors. It identifies the same top sensor as the consensus core in most cases, but assigns it two to three times the normalised weight. This is a known property of LIME's local linear surrogate fitting: when one feature dominates the local prediction surface, the surrogate can allocate nearly all explanatory weight to it, compressing the contributions of secondary features.

PFI produces isolated spikes on sensors that other methods rank as moderate or even low importance. The specific sensor varies by configuration, s8 on FD002, s13 on the all-data model, but the pattern is consistent. These spikes likely reflect individual engines or narrow operating regimes where permuting that sensor causes disproportionate error, inflating the global average.

CF does not produce stable or interpretable rankings in this setting. Its results vary substantially across configurations and rarely agree with the consensus of the other methods. As discussed in the verification section, counterfactual search is sensitive to the local geometry of the prediction surface and to the permitted perturbation ranges. On a 1D-CNN with 13 sensor channels and complex temporal interactions, the search space appears large enough that the results are dominated by search dynamics rather than meaningful feature effects.

The transition from single-dataset to multi-dataset training reveals a shift in which sensors dominate. On FD001 alone, s9 leads. On FD002 and multi-dataset configuration, s11 takes over. This shift is discussed further in the context of the underlying physical degradation mechanisms in Section 5.7.

5.3.4 Additional configurations: FD004 and Cross-Complexity Pairing

The previous comparisons covered FD001 (simplest), FD002(multi-condition), a pairwise combination and the full dataset. To round out the picture, this section looks at FD004, the most complex subset, with six operating conditions and two fault modes, and the FD001-FD004 pairing, which puts the simplest and most complex subset together in a single training set. On

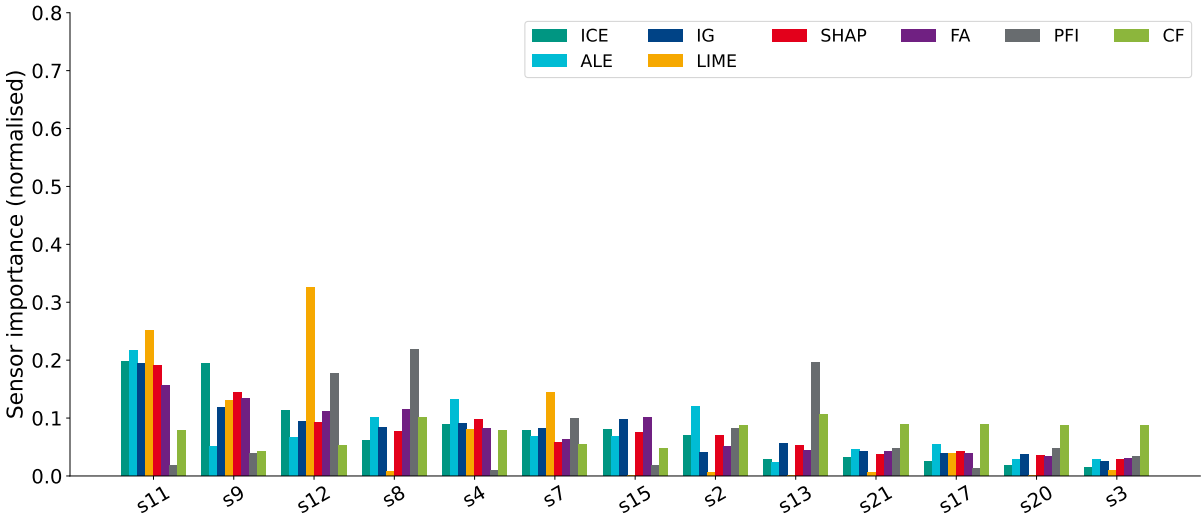

Figure 5.10: L1-normalised sensor importance across all eight methods for the single-FD004 model. Sensors are ordered by decreasing mean importance.

FD004 alone, s11 leads for the consensus methods (IG, SHAP, FA, ALE), just as it did on FD002. That is not surprising, both share six operating conditions and s11 appears to carry the most reliable degradation signal when operating-point variability is high. Sensors s9 and s12 hold the second tier. What is worth noting is that FD004's two fault modes create a richer training set where no single sensor's permutation causes the kind of catastrophic error spikes seen on FD002. CF remains diffuse, with no strong preference for any particular sensor.

Comparing FD004 single against FD001 single directly, the contrast is stark. On FD001, s9 dominates and the consensus methods assign it roughly double the importance of any other sensors. The model trained on the more complex data has learned to spread its reliance across multiple signals rather than leaning heavily on one. A pattern that makes sense when operating conditions vary, since any single sensor's value becomes a less clean indicator of degradation state. The pairwise FD001-FD004 model is interesting because it forces the network to reconcile the simplest and most complex subsets. The resulting ranking is close to what the all-data model produces: s11, s9 and s12 as the top three, with s4 and s15 following. The consensus methods agree more tightly here than in either single-subset case. This suggests that exposure to both extremes of the complexity spectrum is enough to stabilise the learned feature representation, adding the intermediate subsets (FD002, FD003) contributes less additional information than one might expect. LIME continues to pile importance onto s11 alone and PFI deviates on different sensors than it did on the single models, but neither disrupts the consensus top three.

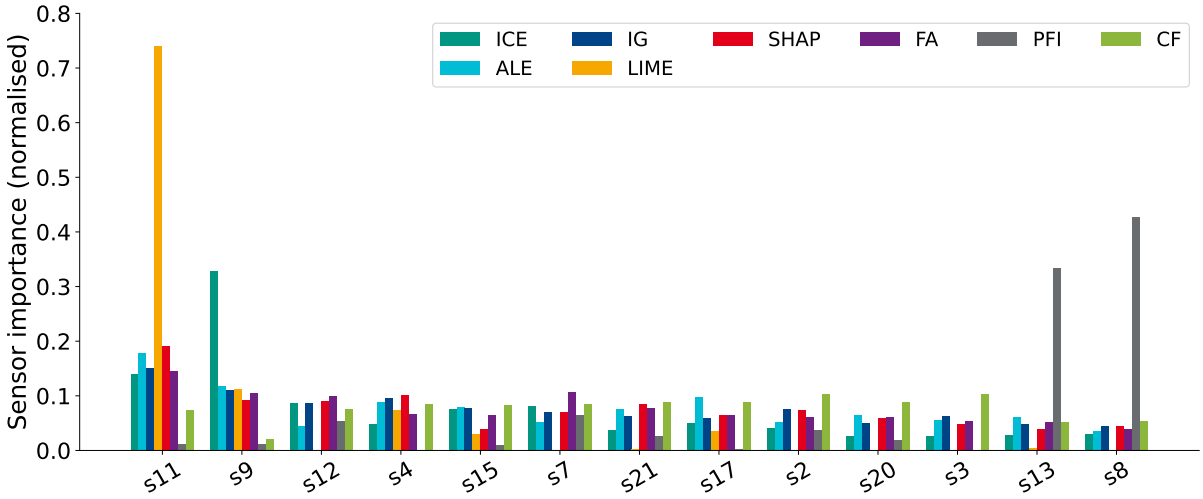

Figure 5.11: L1-normalised sensor importance across all eight methods for the pairwise FD001–FD004 model. Sensors are ordered by decreasing mean importance.

Comparative plots for all remaining model configurations, including single-FD003 and the pairwise combinations FD001-FD002, FD001-FD003, FD002-FD003, FD002-FD004 and FD003-FD004, are provided in Appendix A.2.

5.4 Top-3 Confidence Scoring and Stability Analysis

The previous sections described sensor rankings one configuration at a time. That is useful for understanding how training data affects what the model learns, but it does not answer the aggregate question: across all models and all methods, which sensors come out on top most consistently?

5.4.1 Confidence Scoring

To quantify this, the three highest-ranked sensors from each method on each model configuration receive a weighted score: 3 points for first place, 2 for second, 1 for third. Summing across all configurations and all eight methods yields the aggregate confidence score in Sensor s11 leads with 145 points, more than double the next sensor, s9, at 69. Sensor s12 follows closely at 67. The gap between the top three and the rest is substantial. A middle group, s8 (40), s7 (38), s13 (38), s15(38), s4 (36), clusters tightly between 36 and 40 points. These sensors appear in the top three for some methods on some configurations, but without the consistency of s11, s9 or s12. Below 30 points, sensors appear in the top three only sporadically and s20 (3) and s21 (1) almost never make it.

The dominance of s11 is not driven by any single method or configuration. Even if PFI and CF, the two methods that showed the most erratic behaviour, are mentally excluded, s11 still accumulates the most top-3 appearances from the remaining six methods alone. It ranks first or

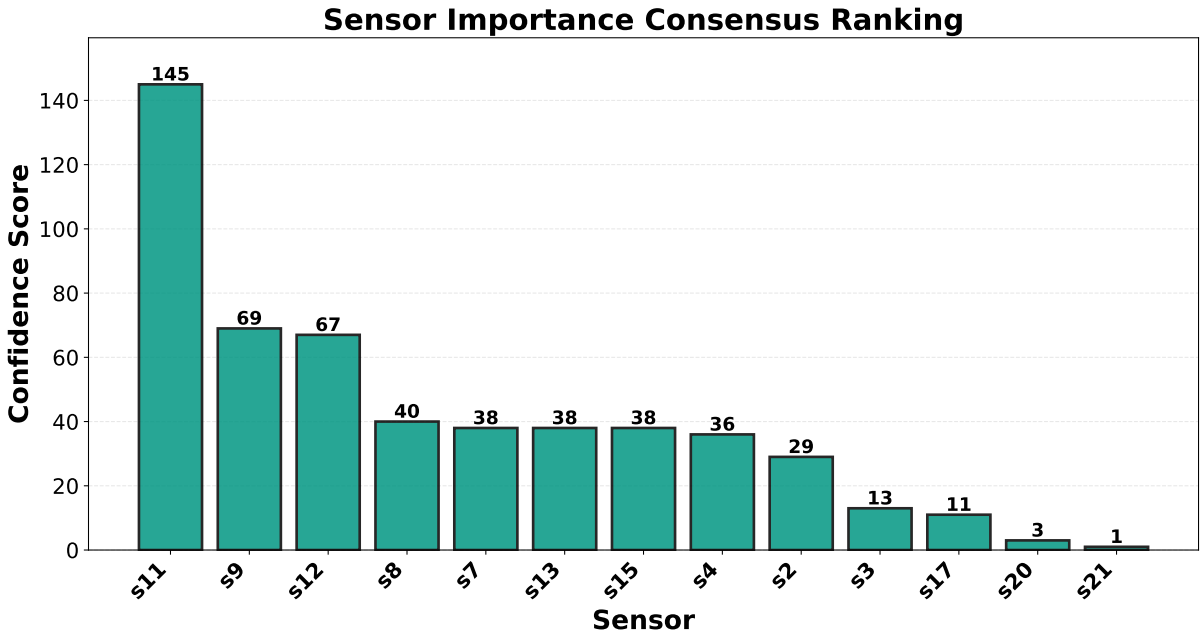

Figure 5.12: Aggregated top-3 confidence scores across all model configurations and feature importance methods. Scoring: 3 points for rank 1, 2 for rank 2, 1 for rank 3, summed over all configurations and methods.

second on every multi-condition model and remains in the top three even on FD001, where s9 takes the lead. That breadth of support across different training conditions is what separates s11 from sensors like s9 or s12, which lead on specific subsets but fade on others.

5.4.2 Mean Importance and Variability

The confidence score captures how often a sensor ranks highly, but not how large its importance is when it does. A sensor could land in the top three with a normalised importance of 0.05 every time and still accumulate a respectable score. Figure 5.13 adds the magnitude perspective. Sensor s11 has the highest mean importance at approximately 0.15. Sensors s12, s9, s7 and s15 follow in the range of 0.08 - 0.11. The ordering in this middle group does not perfectly match the confidence ranking, s12 slightly outperforms s9 in mean magnitude, even though s9 scored marginally higher on the confidence chart. The differences arises because s9 lands in the top three more frequently (accumulating ranking points) while s12 sometimes receives a higher absolute importance value when it does appear.

The error bars tell an important part of the story. Every sensor's standard deviation is large, typically spanning from near zero to twice the mean. Two sources drive this variability. The first is inter-method disagreement: PFI and CF produce outlier values that pull the distribution wider. The second is inter-configuration variation: a sensor that dominates on FD001 may drop to mid-tier on FD004. Sensor s11 has the largest absolute standard deviation but also the highest mean, so its coefficient of variation is comparable to the other top sensors. Sensors s8 and s4

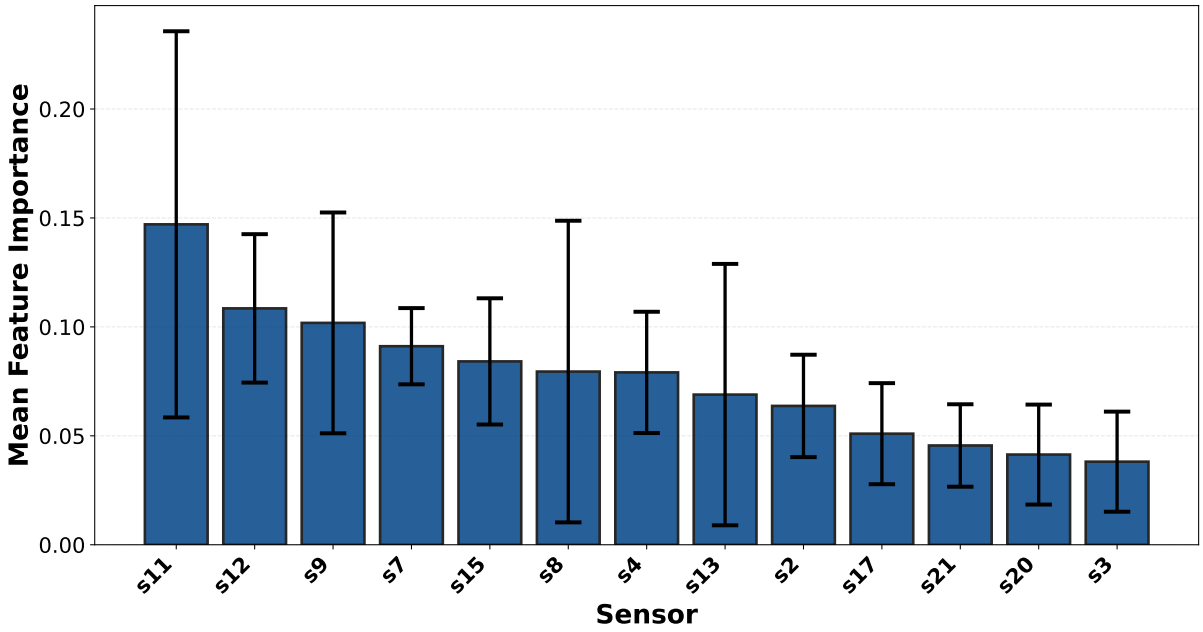

Figure 5.13: Mean L1-normalised sensor importance across all model configurations and methods. Error bars show one standard deviation. Sensors are ordered by decreasing mean importance.

have error bars that extend noticeably low, consistent with configurations where they receive near-zero importance from the consensus methods but are spiked by PFI.

The two analyses converge on the same conclusion: s11 is the most important sensor by both ranking frequency and average magnitude. Sensors s9 and s12 are reliable second-tier features. Beyond those three, the rankings are sensitive to which method and which training configuration is used, which is precisely the kind of instability that a multi-method comparison is designed to reveal.

5.4.3 Cross-Configuration Ranking Agreement

The confidence score and mean importance show which sensors end up on top in aggregate, but they do not quantify how consistent the rankings are *within* a given method across different training configurations. A method might always put s11 first but reorder the rest unpredictably, or it might produce nearly identical rankings regardless of whether it sees FD001 alone or all four subsets.

To measure this, the Spearman rank correlation between the sensor importance vectors was computed for every pair of model configurations, separately for each method. The mean off-diagonal correlation, averaged across all configuration pairs, gives a single stability number per method. Figure 5.14 shows the result.

IG and FA lead with mean correlations of 0.626 and 0.614, the two methods whose rankings shift least when the model is retrained on a different subset or combination of subsets. SHAP

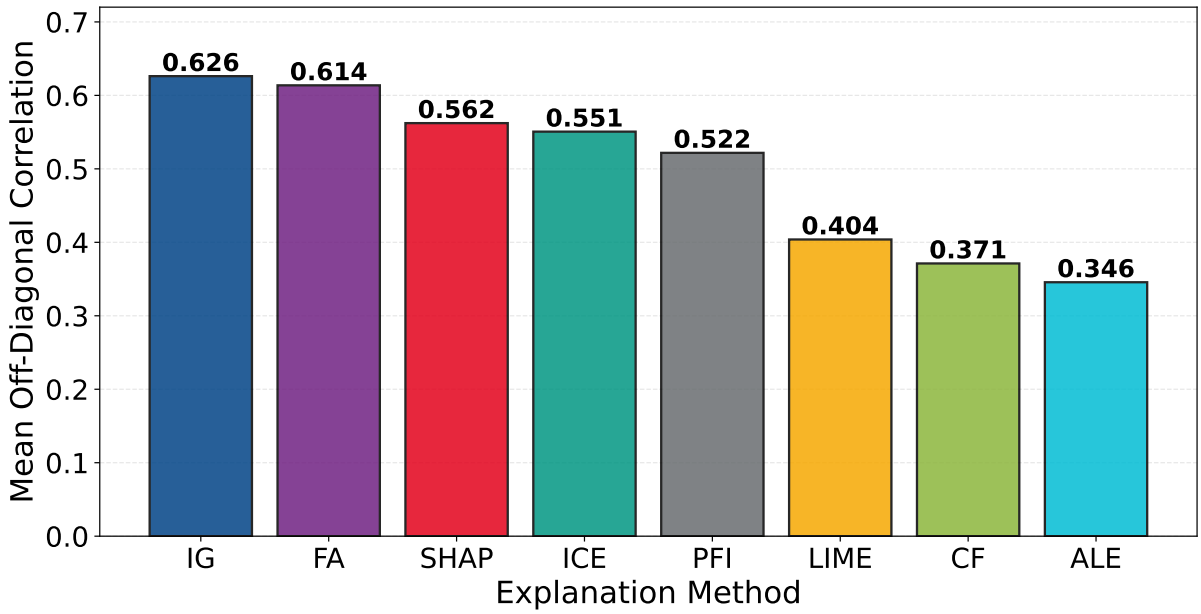

Figure 5.14: Mean off-diagonal Spearman rank correlation across model configurations for each feature importance method. Higher values indicate more consistent sensor rankings when the training data changes.

and ICE follow at 0.562 and 0.511. PFI, despite its tendency to produce magnitude spikes, achieves a moderate 0.522: the spikes inflate individual sensor scores but do not completely scramble the overall ranking order.

The lower end is more surprising. LIME (0.404) and CF (0.371) were expected to score poorly based on the erratic behaviour observed in the per-model comparisons. But ALE lands lowest of all at 0.346, which is unexpected given that ALE consistently agreed with the consensus core on any *individual* model. The explanation lies in what ALE actually measures. It computes local prediction differences within narrow bins of feature values, and those local slopes are sensitive to the specific model being analysed. When the training data changes and the model's internal parameters shift, the local prediction surface rearranges and ALE's bin-level effects can reorder substantially even if the broad ranking of top sensors stays similar. IG and FA, by contrast, accumulate attributions along gradients or through channel-level ablation, which depends more on the model's global structure and less on local surface geometry. That makes them more robust across configurations.

The practical implication is that IG and FA are the most reliable methods for producing sensor rankings that generalise across different training setups. For a single set of "most important sensors" that holds regardless of how the model was trained, IG and FA would be the methods to rely on. SHAP and ICE are reasonable alternatives. LIME, CF and ALE are less dependable for cross-configuration comparisons, even though ALE performs well when applied to a single fixed model.

The individual correlation matrices for each method are provided in Appendix **??**. As an example, the ICE correlation matrix (Figure 5.15) shows values ranging from -0.06 (single-FD003 vs

pairwise FD001-FD003) to 0.93 (pairwise FD002-FD004 vs FD003-FD004), illustrating both the extremes and the general trend. Models that share similar training conditions, particularly those involving multi-conditions subsets, tend to produce more correlated rankings than models trained on different single-condition subsets.

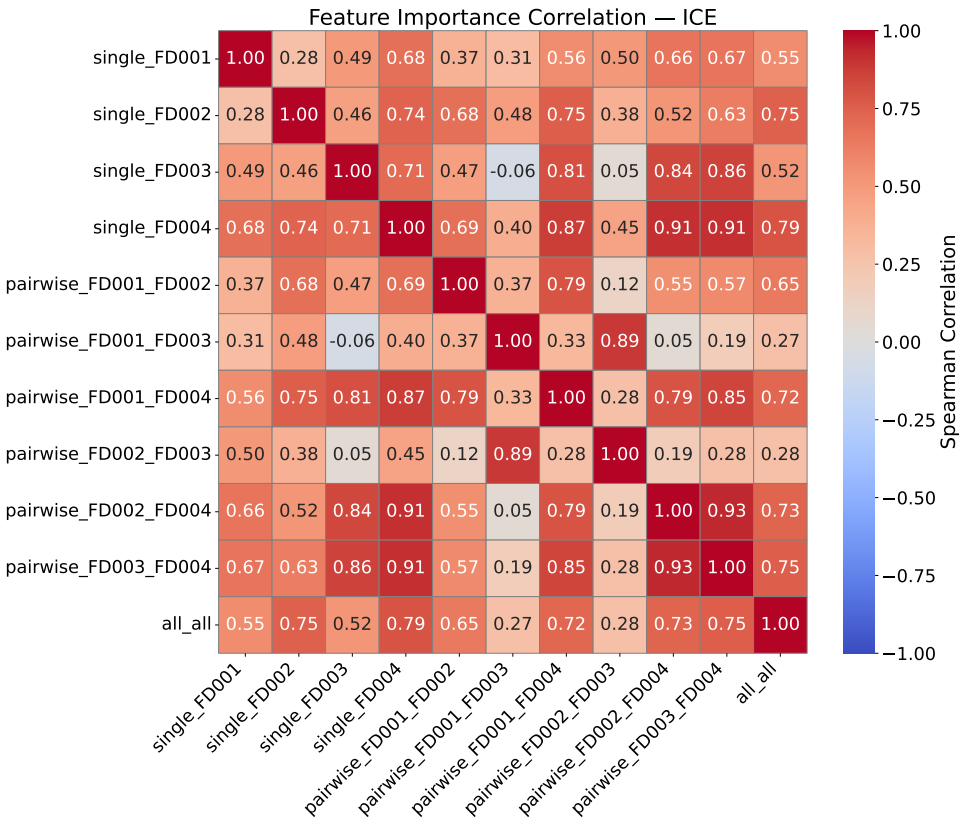

Figure 5.15: Spearman rank correlation of sensor importance rankings across model configurations for ICE. Red indicates high agreement, blue indicates disagreement. Complete matrices for all eight methods are provided in Appendix **??**.

5.5 Temporal Distribution of Feature Importance

The analyses above aggregate importance across the full sliding window. But the 1D-CNN receives 70 time steps as input and there is no a priori reason to assume every time step matters equally. Four of the eight methods produce attributions resolved at the individual time step level, Integrated Gradients, Feature Ablation, LIME and Counterfactual Explanations, and their heatmaps reveal a clear temporal pattern. Integrated Gradients shows a clear gradient from dark (low importance) in the early time steps to bright (high importance) in the final 10-15 steps. The concentration is strongest for sensors s9 and s8, the same sensors that rank highest in the aggregated bar charts. The pattern is not binary (all-dark then all-bright) but progressive: importance builds gradually across the second half of the window and peaks sharply at the end.

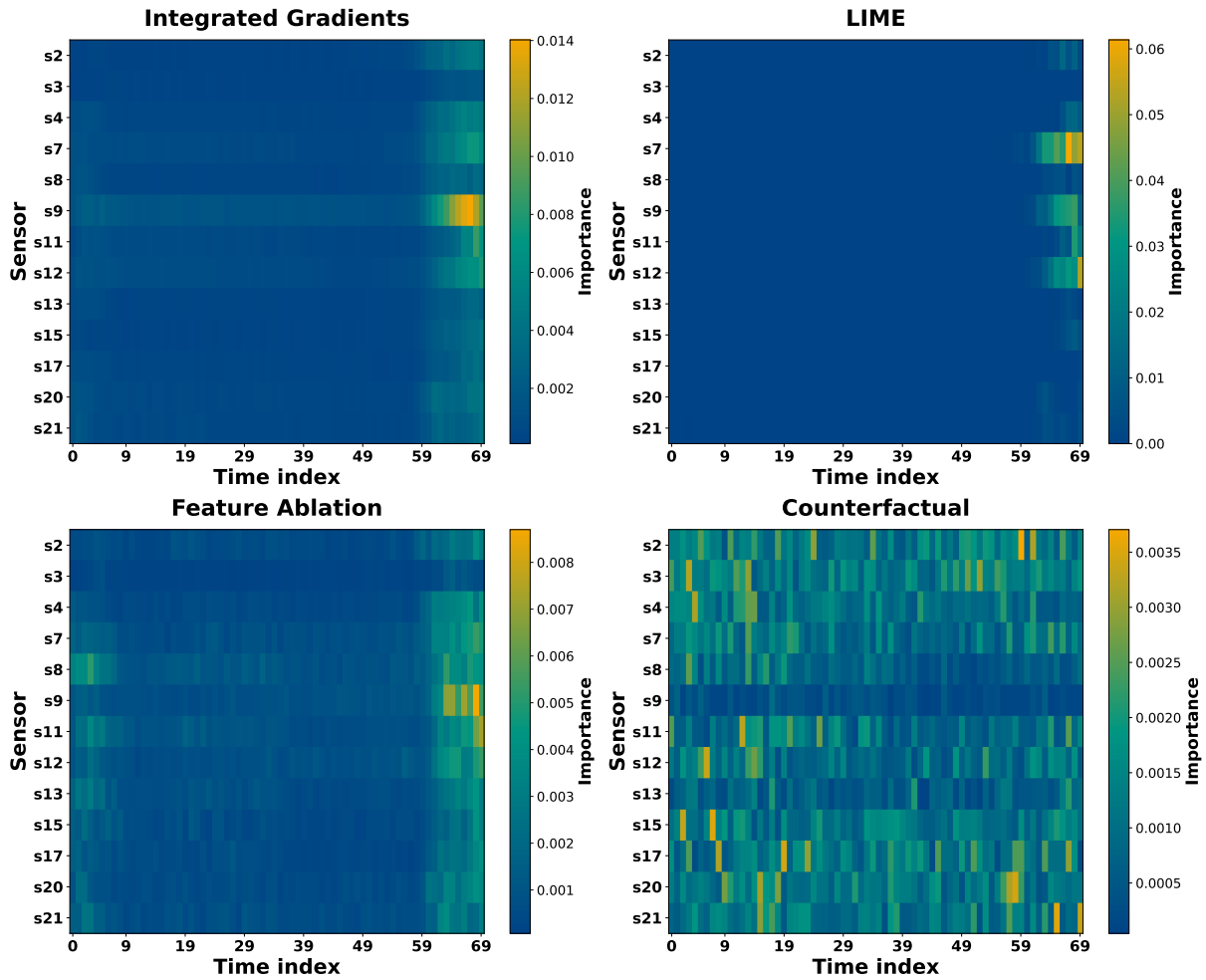

Figure 5.16: Sensor-by-time importance heatmaps for four methods on the single-FD001 model. Horizontal axis: time index within the sliding window (0 = earliest, 69 = most recent). Vertical axis: sensors. Brighter colours indicate higher importance.

This suggests the model is not simply ignoring early time steps; it assigns them some weight, but the most recent readings carry the prediction.

Feature Ablation shows a similar pattern, though slightly less concentrated. The bright region extends back further, perhaps 15-20 time steps from the end and lights up sensors s2 and s9 in addition to s8. The broader temporal spread could reflect the difference between how the two methods work: IG traces gradients through the network, which can sharpen around the most influential time steps, while FA zeroes out entire channels and measures the prediction change, which may pick up on the cumulative effect of a sensor over a wider window.

LIME produces the most concentrated heatmap of the three. Nearly all importance sits in the final 5-10 time steps and only on sensors s7 and s4. The rest of the heatmap is essentially blank. This extreme concentration is consistent with LIME's tendency to over-focus on the locally most influential features, here, the most influential sensor at the most influential time step captures almost everything.

Counterfactual Explanations are the outlier. The heatmap is noisy, with scattered bright spots across the full window and across all sensors. There is no temporal concentration. This is consistent with everything observed about CF in the previous sections: the method's search dynamics spread perturbations across the entire input space without converging on a consistent pattern. Where IG, FA and LIME all agree that the late window matters most, CF provides no such signal.

The late-window concentration is a genuine finding about the model, not an artefact of the explanation methods. Three independently designed methods, one gradient-based, one ablation based, one surrogate based, converge on the same temporal pattern. The model relies most heavily on the most recent sensor readings when predicting RUL. Earlier readings in the window contribute some context, but the current snapshot dominates. Whether this pattern holds across all model configurations or shifts as training data complexity increases is an open question; the heatmaps shown here are from the single-FD001 model and the temporal structure could differ on multi-condition models where operating point transitions create richer temporal dynamics.

5.6 Reduced Sensor Model Performance

The feature importance analysis identified sensors s11, s9 and s12 as the three most consistently important across methods and training configurations. A natural follow-up question is whether the model can maintain its predictive performance using only these three sensors instead of the full set of thirteen. To test this, new models were trained on the same configurations, single-subset, pairwise and all-data, but with only s11, s9 and s12 as input channels. The resulting RMSE and R^2 values were compared against the full-sensor baselines.

5.6.1 All-Data Model

The most striking results come from the model trained on all four subsets. Table 5.1 compares the full-sensor and top-3-sensor models. The reduced model does not just match the full-sensor

Test subset	Full sensors (RMSE)	Top-3 sensors (RMSE)	Top-3 R^2
FD001	10.60	8.85	0.95
FD002	10.13	9.64	0.94
FD003	10.56	7.95	0.96
FD004	11.04	9.11	0.95

Table 5.1: RMSE comparison between the full 13-sensor model and the top-3 sensor model (s11, s9, s12), both trained on all four subsets.

model, it outperforms it on every subset. RMSE drops by 10-25% and R^2 values sit at 0.94-0.96 across the board. This is a counterintuitive result at first glance: fewer inputs, better predictions. But it has a straightforward explanation. The ten sensors removed from the input are, according

to the importance analysis, carrying minimal predictive signal. What they do contribute is noise, variability that the CNN must learn to ignore. With fewer input channels, the model has a simpler learning task and fewer parameters to fit, which reduces overfitting and improves generalisation. The training data in the all-data configuration is large enough (all four subsets combined) that the model does not need the additional channels to capture degradation patterns.

5.6.2 Single-Subset Models

The single-subset results tell a very different story. When the model is trained on only one subset and tested on the same subset, the top-3 reduction causes severe performance degradation. The

Configuration	Full sensors (RMSE)	Top-3 sensors (RMSE)
FD001	10.72	100.19
FD002	11.78	43.08
FD003	9.81	38.68
FD004	12.00	121.48

Table 5.2: RMSE comparison between the full 13-sensor model and the top-3 sensor model (s11, s9, s12) for single-subset training configurations. Each model is tested on the same subset it was trained on.

RMSE increases by factors of 3-10. On FD001, the model goes from an RMSE of 10.72 to over 100, essentially useless predictions. FD004 is similarly catastrophic. FD002 and FD003 fare somewhat better but still degrade by a factor of 3-4.

The difference between the all-data and single-subset outcomes makes sense when considered alongside the importance analysis. On FD001 alone, the model relied heavily on s9, with s7, s12 and s2 as important secondary sensors. Removing those secondary sensors, which are not in the top-3 overall ranking but are critical for the FD001-specific model, destroys the prediction. The top-3 sensors (s11, s9, s12) were identified as the most important across all configurations, but that aggregate ranking does not reflect the sensor dependencies of any single-subset model. A model trained only on FD001 never learned to rely on s11 as heavily as the multi-condition models did, so including s11 while removing s7 is a net loss.

5.6.3 Pairwise Models

The pairwise models fall between the two extremes. Most pairwise top-3 models achieve RMSE values in the 30-40 range when tested on the subsets they were trained on, compared to 10-15 for the full-sensor versions. The degradation is significant but not catastrophic and the models still produce physically meaningful predictions. The exception is FD003-FD004 trained and tested on FD001 or FD002, where RMSE reaches 60, but these are cross-subset evaluations where even the full-sensor models struggle. Full pairwise heatmaps are provided in Appendix A.3.

The reduced sensor experiment reveals a clear interaction between training data diversity and sensor selection. When the model has access to varied training data covering multiple operating conditions and fault modes, three well-chosen sensors are enough, and in fact perform better than the full set, likely due to reduced noise and lower model complexity. When training data is limited to a single subset, the same three sensors are insufficient because the model's learned feature dependencies do not align with the aggregate importance ranking. The practical implication is that sensor reduction based on feature importance analysis is viable, but only when paired with sufficiently diverse training data.

5.7 Discussion

The results presented in this chapter address three questions: which sensors does the 1D-CNN rely on for RUL prediction, how consistently do different explanation methods answer that question and what practical consequences follow from the answers.

5.7.1 Sensor Importance and Physical Interpretation

Across all training configurations and explanation methods, three sensors emerge as the most consistently important: s11, s9 and s12. Sensor s11, static pressure at the HPC outlet, leads by a substantial margin in both the confidence scoring and the mean importance analysis. This is physically plausible. HPC outlet pressure is directly affected by compressor degradation: as blade erosion, fouling or tip clearance increases, the compressor's pressure ratio drops and s11 captures that change. Under multiple operating conditions, s11 remains a reliable degradation indicator because the relationship between compressor health and outlet pressure holds across different altitude-speed-throttle combinations, even though the absolute pressure level varies with the operating point.

Sensor s9, core speed, dominates on FD001 but drops to a secondary role on multi-condition subsets. This too is physically consistent. Under a single operating condition, the control system adjusts fuel flow, which shifts the spool speed. Under multiple operating conditions, core speed varies substantially with the operating point itself, making it a noisier indicator of degradation. The model trained on FD001 can afford to lean heavily on s9 because operating-point variability is absent. Multi-condition models cannot.

Sensor s12, bleed enthalpy at the HPC outlet, correlates with both temperature and pressure at the compressor exit and picks up degradation effects through both pathways. Its consistent presence in the top three across configurations suggests it captures complementary information to s11.

The shift in sensor dominance from s9 (single-condition) to s11 (multi-condition) is one of the clearer findings. It means that a feature importance analysis conducted on a single-condition model would lead to different conclusions about sensor priority than one conducted on a multi-condition model, a concrete example of why training data composition matters for interpretability, nt just for prediction accuracy.

5.7.2 Method Reliability

Not all eight methods proved equally useful. The cross-configuration Spearman analysis revealed a hierarchy.

Integrated Gradients and Feature Ablation are the most stable, producing consistent sensor rankings regardless of which data the model was trained on (mean correlations of 0.626 and 0.614). They agree with each other and with SHAP on the identity of the top sensors and their magnitude estimates are similar. For practical use, choosing which sensors to monitor, or deciding which to remove, these two methods offer the most dependable guidance.

SHAP and ICE are close behind in stability (0.562 and 0.551) and broadly agree with IG and FA on the top rankings. SHAP adds the advantage of directional and instance-level information thorugh beeswarm and waterfall plots, which IG and FA do not natively provide. SHAP remains the most informative single method, for understanding how a sensor influences the prediction, not just how much.

LIME consistently identifies the correct top sensor but over-concentrates importance, assigning it two to three times the weight that other methods give it. This makes it useful for quick identification of the most dominant feature but unreliable for ranking secondary sensors or for quantitative importance estimates.

PFI produces magnitude distortions, isolated spikes on sensors that other methods rank as moderate, and its cross-configuration stability is only moderate (0.522). The spikes do not consistently appear on the same sensor, which limits trust in PFI as a standalone tool for this application.

ALE agrees well with the consensus core on any individual model but has the lowest cross-configuration stability of all methods (0.346). This was the most surprising finding. The likely explanation is that ALE's bin-level local slopes are sensitive to the specific model parameters and those parameters shift when the training data changes. ALE is useful for understanding a specific model but not for drawing conclusions that should generalise across training configurations.

Counterfactual Explanations were the least reliable throughout, in the verification test, in the per-model comparisons, in the temporal heatmaps and in the cross-configuration stability analysis. The method's search-based approach appears poorly suited to this setting, where 13 input channels and 70 time steps create a perturbation space too large for the search to converge reliably. This does not mean counterfactual explanations are inherently flawed, they are designed for a different purpose (generating actionable "what-if" scenarios) and may perform better on simpler or lower-dimensional models.

5.7.3 Temporal Concentration

The temporal heatmaps revealed that the model concentrates its attention on the final 10-15 time steps of the 70-step sliding window. Three independent methods, IG, FA and LIME, converge

on this pattern. The model is primarily using the most recent sensor readings to form its RUL prediction, with earlier readings contributing context but not dominating.

This has practical implications. If the model's predictive power is concentrated in the most recent readings, shorter sliding windows might achieve comparable accuracy at lower computational cost. It also suggests that the model is functioning more as a snapshot-based estimator with some temporal context than as a full trajectory learner, a distinction that could inform architecture choices for future work.

5.7.4 Reduced Sensor Viability

The reduced sensor experiment produced the chapter's most practical finding. Three sensors, s11, s9, s12, are sufficient to match or exceed the full sensor model's performance, but only when the model is trained on diverse data spanning multiple operating conditions and fault modes. On single-subset models, the same reduction is catatrophic.

This result has a direct engineering implication. If a prognostic system is deployed with a model trained on sufficiently varied operational data, monitoring could be simplified to three sensors without sacrificing accuracy. But a model developed and validated on narrow operating data should not be reduced in the same way, its learned feature dependencies may differ substantially from the aggregate importance ranking.

5.7.5 Limitation

Several limitations need to be noted. The analysis is conducted on a single model architecture, a 1D-CNN, and the results may not transfer directly to recurrent or attention-based architectures. The C-MAPSS dataset, while the standard benchmark, is simulated data; real-world sensor data contains additionoal noise, calibration drift and failure modes not captured in the simulation. The counterfactual explanations underperformed across the board, but this may reflect the specific implementation and parameter choices (target range, perturbation limit) rather than a fundamental limitation of the method. The temporal heatmaps were shown for a single configuration (FD001 single); whether the late-window concentration pattern holds under all training setups was not systematically evaluated. Finally, the top-3 sensor reduction is a conservative test, intermediate reduction (top-5, top-7) were not explored and might offer a better accuracy-simplicity tradeoff on single-subset models.

6 Conclusion and Outlook

6.1 Conclusion

This thesis applied eight feature importance methods to a 1D-CNN trained for remaining useful life prediction on the C-MAPSS turbofan engine dataset. The goal was not to improve prediction accuracy, the model itself was fixed, but to understand what it had learned and to evaluate whether different explanation methods agree on that understanding.

They do not fully agree and the disagreements turned out to be as informative as the agreements.

Three sensors, s11, s9 and s12, emerged as the most important across all methods and training configurations. Sensor s11 led by a wide margin in both ranking frequency and average magnitude, with s9 dominant on single-condition subsets and s11 taking over when operating-point variability increased. This shift is physically consistent with the role of HPC outlet pressure as a direct indicator of compressor health and the confounding effect of varying operating conditions on core speed measurements.

The eight methods split into a clear reliability hierarchy. Integrated Gradients and Feature Ablation produced the most stable rankings across training configurations, followed by SHAP and ICE. LIME identified the correct top sensor but over-concentrated its importance estimates. PFI preserved broad ranking order but distorted magnitudes through isolated spikes. ALE, despite agreeing with the consensus on individual models, proved the least stable across configurations, and unexpected result that highlights the sensitivity of bin-level local effects to changes in model parameters. Counterfactual Explanations were unreliable throughout, producing inconsistent rankings, noisy temporal attributions and search-dominated results that did not converge to stable feature importance estimates in this high-dimensional setting.

The temporal analysis revealed that the model concentrates its predictive attention on the final 10-15 time steps of the 70-step sliding window. Three independent methods, gradient-based, ablation-based and surrogate-based, converged on this pattern, lending confidence that it reflects the model's actual behaviour rather than an artefact of any single explanation technique.

The reduced sensor experiment produced the most directly applicable finding. A model trained on all four C-MAPSS subsets using only the top-3 sensors outperformed the full 13-sensor model on every test subset, with RMSE reductions of 10-25% and R^2 values above 0.94. The same reduction applied to single-subset models caused catastrophic performance loss. Sensor selection based on aggregate feature importance is viable, but only when paired with training data that spans a sufficient range of operating conditions and fault modes.

Taken together, the results make a case for multi-method explainability analysis as standard practice in data-driven prognostics. Relying on a single explanation method, as most existing studies do, risks mistaking that method's particular perspective for a complete picture of the model's behaviour. Comparing multiple methods exposes where conclusions are robust and where they depend on the choice of tool.

6.2 Outlook

Several directions follow naturally from this work.

The most immediate is extending the analysis to other model architectures. The 1D-CNN studied here is relatively simple, no recurrence, no attention, no gating. LSTM-based, Transformer-based and hybrid CNN-LSTM models are widely used for RUL prediction and may distribute feature importance differently. Whether IG and FA remain the most stable methods on these architectures and whether the late-window temporal concentration persists in model with explicit temporal memory, are open questions.

The reduced sensor finding invites further investigation. This thesis tested only the top-3 reduction. Intermediate levels, top-5, top-7, top-9, would map out the accuracy-simplicity tradeoff more precisely and might identify a point where single-subset models can tolerate sensor reduction without catastrophic loss. Similarly, sensor selection could be made configuration-specific rather than aggregate: choosing sensors based on the model's own importance ranking rather than the global consensus might improve reduced-sensor performance on narrow training sets.

The temporal concentration result suggests that shorter sliding windows could achieve comparable prediction accuracy at lower computational cost. A systematic sweep of window lengths, combined with the temporal attribution analysis, would quantify how much of the window is actually needed and whether the optimal length varies across subsets or training configurations.

Counterfactual Explanations underperformed in this thesis, but the method was applied in its most general form, searching over all sensors and all time steps simultaneously. Constraining the search to physically meaningful perturbation directions, or reducing the dimensionality of the input before generating counterfactuals, might produce more interpretable results. This is worth exploring separately rather than writing off the method based on its performance here.

Finally, the analysis was conducted entirely on simulated data. Applying the same multi-method comparison to real-world engine monitoring data, with sensor drift, calibration errors, missing values and failure modes not represented in C-MAPSS, would test whether the method reliability hierarchy and the sensor importance rankings transfer to operational conditions. That step is necessary before the findings of this thesis can inform actual maintenance decisions.

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Appendix

A.1 Individual Results

A.2 Comparative Results

A.3 Reduced Sensor